NOTICE OF INTENT

This document provides information concerning the major tasks and typical responsibilities associated with a frame 6001B Hot Gas Path Inspection. It describes the overall work scope, but does not detail specific methods. Its primary purpose is to provide an overview of a typical inspection process. Design changes subsequent to the issue date of this document may not be reflected in this guide.

This document does not purport to cover all aspects of a particular inspection. Unit-specific details are provided in the document and drawing packages supplied with each requisition, and all sketches and descriptions are to be considered "Typical." This document is property of General Electric and is considered Company Proprietary Information.

HEALTH AND SAFETY REQUIREMENTS

The GI&FS Environmental Health & Safety Manual (EHSM) outlines the requirements that must be followed on the job site during an installation or maintenance outage.

Safety, health, and protection of the environment must be an integral part of all business operations, design, construction and start-up, and includes EHS planning prior to and during work to anticipate and minimize hazards and incidents. Every reasonable effort must be made to create and maintain a safe and healthy work environment at each customer site.

Where working conditions are not covered by these safety requirements and procedures, or when a requirement or a procedure is not completely understood, obtain specific instructions or clarification from manager, supervisor or EHS before proceeding with the work.

INTRODUCTION

This Maintenance Procedure currently exists as a part of a Gas Turbine Service Manual, and is intended to establish a consistent approach to performing flange-to-flange maintenance inspections for an MS6001B gas turbine. This procedure consolidates actual field experience and other maintenance related information normally available only from widely diverse sources. As maintenance experience is gained, this procedure will be supplemented with additional information and descriptions of the work to be performed. This article currently provides inspection information for Hot Gas Path Inspection workscope. The procedures described in the text do not supersede or replace specific product department documentation.

REFERENCE DOCUMENTS

No.	Description
EM 5202	Gas Turbine Mechanical Support and Jacking Instructions

HOT GAS PATH INSPECTION

Caution

Before proceeding with a hot gas path inspection ensure that the gas turbine electrical power is tagged out, CO2 system is deactivated, liquid fuel system is purged, deactivated and/or the gas supply is blanked off.

DISASSEMBLY

OPERATION 1 - PERFORM COMBUSTION INSPECTION DISASSEMBLY OPERATIONS NUMBER 1 THROUGH NUMBER 16

NOTE

Refer to Combustion Inspection Section 4A of this Service Manual for disassembly procedures for the following operations.

1. Prepare turbine compartment roof for removal.
2. Remove turbine compartment roof and side panels.
3. Remove liquid fuel lines.
4. Remove atomizing air lines.
5. Remove gas fuel lines (if applicable).
6. Remove steam injection lines (if applicable).
7. Remove water injection lines (if applicable).
8. Remove liquid fuel check valves.
9. Remove false start drain tubing and manifold header.
10. Remove fuel nozzles.
11. Remove flame detectors and spark plugs.
12. Remove 11th stage cooling and sealing air lines.
13. Unbolt and open up combustion outer casing covers.
14. Remove crossfire tube retainers, crossfire tubes, combustion liners and forward flow sleeves.

15. Remove the atomizing air extraction manifold from the upper half turbine casing. In addition to the extraction manifold, remove the upper half inlet deicing piping, if applicable, and the upper half exhaust frame cooling piping. Discard all gaskets and cover all pipe openings. Identify the bolting for reassembly.

16. Remove outer combustion casings.

OPERATION 2 - REMOVE EXHAUST AND INLET PLENUM ACCESS PANELS

1. On straight up exhaust configurations, unbolt the removable exhaust duct transition piece panel on the forward side of the transition piece.

2. On side exhaust configurations remove any architectural lagging and support members which may interfere with turbine casing removal.



Before the removable panel on the forward wall of the units exhaust plenum can be removed, the upper section (right side looking with air flow) of the 11th stage extraction piping and compressor bleed valve (33CB-2) must be removed.

3. Disconnect the tubing runs attached to the head of the bleed valve and the tubing runs attached to the extraction piping at both ends. Remove the "U" bracket that supports the actuating cylinder, loosen the slip joint clamp and unbolt the extraction line from the compressor casing.

4. Rig to remove the piping, slip the piping out of the expansion joint and lift the piping clear of the unit. Set the piping on suitable material to protect the valve and tubing runs. Cover all openings on piping and compressor casing.

5. Remove the forward upper half segment clamp bars and the flex seal plates between the exhaust hood frame and exhaust plenum.

6. Unbolt the forward exhaust plenum removable access panel. Rig to lift the panel.



The compressor bleed valve discharge pipe, insulating packs and the compressor bleed valve diffuser box are part of this panel and should not have to be removed.

7. Install a length of 4 inch pipe or timber and 10 ton mechanical jack across the horizontal opening of the plenum and spread the plenum sufficiently enough to remove the access panel. Lift panel clear of the unit.

8. Unbolt and remove the inlet plenum access door located in the lower right hand section of the plenum, looking with air flow.

OPERATION 3 - TAKE INITIAL COMPRESSOR AND TURBINE ROTOR POSITIONING CHECKS

Rotor positioning checks must be done with all casings bolted in place and unit supported on its own supports.



Do not confuse clearanceometer holes and probe holes. Clearance meter holes are too small for a depth micrometer. Use probe holes for taking tip clearances on turbine and compressor. Compressor or Turbine probe plugs may become stuck or tight over time based upon material, temperature and torque procedures. If difficulty occurs while removing a plug, a judgment must be made as to whether the reading is necessary. Caution must be maintained if the plug is to be removed by drilling. A bushing or stop block should be used to prevent the drill bit from exceeding the required travel distance X, as seen in HGP-D1.

1. Using random 1st stage compressor blades, take feeler clearance checks at six points: top centerline, bottom centerline, and above and below the horizontal joint on each side. Record clearances on Inspection Form GT7200. It may be necessary to use the hydraulic ratchet to position a random blade tip below the probe holes for the 13th and 17th stage compressor and 1st stage turbine for rotor positioning checks.
2. The 13th stage compressor holes (4 total) are located 7.46 inches aft from the forward face of the compressor discharge casing forward flange.
3. The 17th stage compressor probe holes (4 total) are located 19.96 inches aft from the forward face of the compressor discharge casing forward flange.
4. The first stage turbine probe holes (4 total) are located 27.44 inches aft from the forward face of the turbine casing forward flange.
5. Compressor tip clearances at the 13th and 17th stages are taken by removing the plugs located ten (10) degrees off top and bottom vertical centerline and sixteen (16) degrees above each horizontal joint. A stamped number next to the plug counterbores indicates the thickness of the casing from the bottom of the counterbore to the inside diameter of the casing. Record clearances on Inspection Form GT7200.
6. Turbine casing probe holes exist (4 total) so that the turbine bucket tip clearances can be taken on the first-stage turbine buckets. Record the turbine tip clearances on Inspection Form GT7200. Turbine rotor tip clearance readings are taken as follows:
    a. Remove plug and place an adapter into the machined counterbore of the casing. The adapter will allow the base of the depth micrometer to rest squarely on the casing, thereby allowing the micrometer rod to penetrate the probe hole accurately.

NOTE The adapter can be made from a piece of 0.075 inch diameter pipe approximately two inches long. The ends of the pipe faces should be within 0.0005 inch. See Figure HGP-D1.

Take the depth micrometer and measure to the tip of the blade. This measurement will be the distance from the blade tip to the top of the adapter. Add the adapter length to the thickness stamped on the turbine casing. Subtract this measurement from the depth micrometer measurement reading. The result is tip clearance. See Figure HGP-D1.

Caution **Do not rotate the rotor while in the process of taking tip clearances.**

OPERATION 4 - DISCONNECT WIRING AND CONDUIT IN TURBINE COMPARTMENT AREA (REMOVE ONLY IF IT WILL INTERFERE WITH TURBINE AREA WORK)

1. Disconnect the electrical leads for the ac/dc lighting at the pull box above turbine casing.
2. Pull wiring from conduit and tag for identification.
3. Tag and remove sections of conduit as needed to provide access to turbine work.

OPERATION 5 - PLACE MECHANICAL SUPPORT JACKS UNDER TURBINE CASING

1. Install mechanical support jacks per EM 5202, Gas Turbine Mechanical Support and Jacking Instructions.

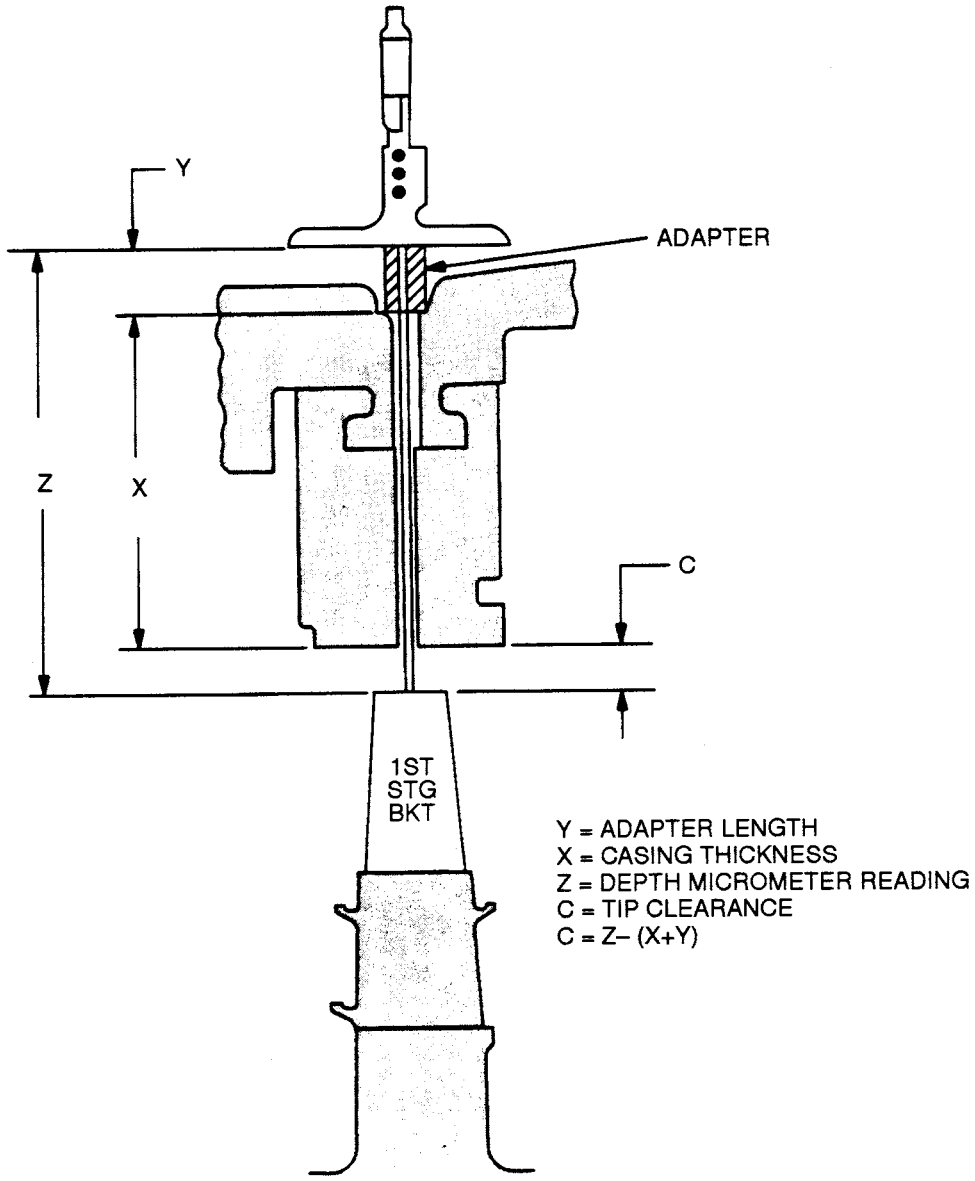


Figure HGP-D1. Rotor Position Probe

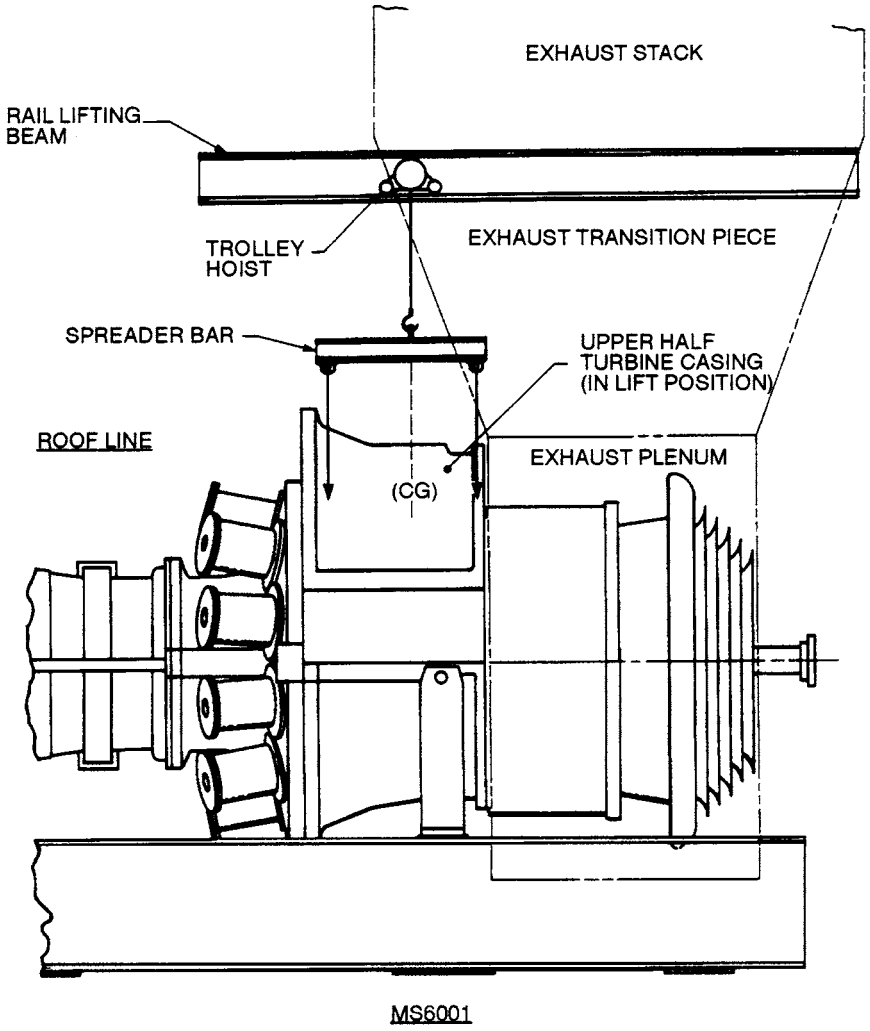


Figure HGP-D2. Turbine Casing Rigging for Removal

OPERATION 6 - REMOVE TURBINE CASING BOLTS AND UPPER HALF FIRST STAGE NOZZLE ECCENTRIC PIN

1. Unbolt the upper half circumferential bolting between the turbine casing and exhaust hood.
2. Unbolt the upper half circumferential bolting between the turbine casing and the compressor discharge casing.
3. Remove the horizontal joint bolting and body-bound bolting. (Horizontal body-bound bolts are removed by driving them upward.)
4. Mark body bound bolts for replacement in same holes.
5. Remove upper half eccentric pin near the top vertical centerline of the turbine casing. Stamp the pin to identify it as the top eccentric pin.

OPERATION 7 - REMOVE UPPER HALF TURBINE CASING (SEE FIGURE HGP-D2)

The lifting beam, trolley hoist and spreader bar are applicable to vertical exhaust stack configurations only. For side exhaust stack configurations use overhead crane and normal rigging.

1. Procure the turbine casing lifting beam and trolley hoist. Install the stops on the forward end of the lift beam and install the trolley hoist. Clamp the trolley hoist in the forward most position on the lift beam and install the aft stops.
2. Rig to lift the beam-hoist assembly with an overhead crane. Lift the beam-hoist and guide it into the forward opening of the exhaust duct transition piece as far as the rigging will allow.
3. Using a chainfall or comealong mounted in the transition piece, slide and guide the lift beam into the mounting slot(s), bolt the lift beam at the slot(s) and at the plate on the beam to the transition piece.
4. Remove the rigging and release the clamp on the trolley hoist.
5. Attach the spreader bar to the trolley hoist hook and position the trolley over the turbine casing.
6. Lubricate and install guide pins and jackbolts. Refer to Standard Practices, Section 2 - Special Tools in the Service Manual.

7. All guide pins are used in the body bound bolt hole provisions. See Figure HGP-D3. Tapped holes (1.75 inch-eight thread) for jacking the turbine casing apart are provided at four locations in the upper and lower halves of the turbine casing horizontal joint bolting flange. These are located as follows on each side of the turbine casing.

   a. First hole aft of the forward vertical flange.

   b. Fifth hole forward of the aft vertical flange.

8. Eight 1.75 inch-eight thread jack bolts will be required. Four bolts are to be used for jacking and four to plug the lower half holes. Care must be taken to avoid the bolts crossing the horizontal joint.

9. Using wire rope slings, rig the two aft lower lugs of the spreader bar to the aft lugs of the turbine casing. Using chainfalls or comealongs and wire rope slings rig the forward lower set of lugs on the spreader bar to the forward lugs on the turbine casing. Position the spreader bar as close to the casing as a safe lifting angle will allow. Rigging should be such that when the wire rope slings are taut, the bottom of the spreader bar should be fairly close to the highest section of the turbine casing. The upper half turbine casing weight with the second and third stage nozzle segments and shrouds is approximately 7300 pounds.

10. Using the jackbolts, jack the upper half casing as far as the jackbolts will permit. Keep casing level at all times while jacking.

11. Take up all slack on the wire rope slings and proceed to lift the turbine casing with the trolley hoist until the turbine casing clears the aft vertical flange of the compressor discharge casing.



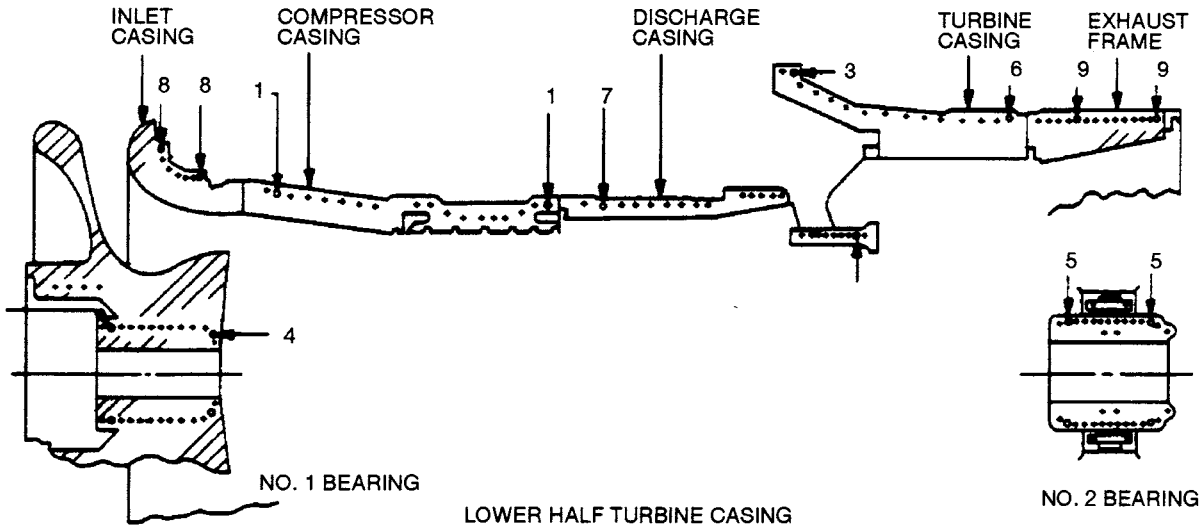
Make certain the casing is lifted vertically and level at all times. Adjust chainfalls or comealongs as needed to locate center of gravity and to level the casing.

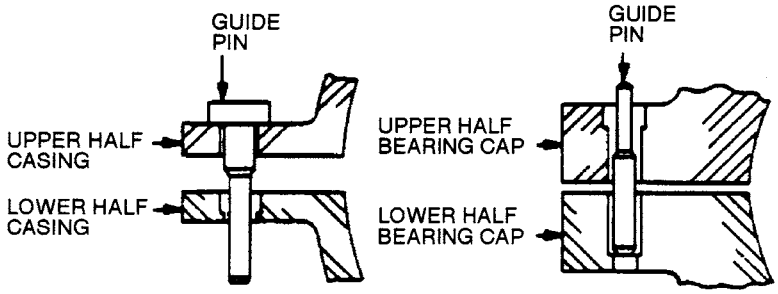
12. Move the trolley hoist to the forward "Stop" on the lifting beam.

13. Using an overhead crane centered precisely with the turbine axis, rig cables to the upper lugs of the spreader bar. Carefully lift the spreader bar with the overhead crane until the rigging between the trolley hoist and spreader bar can be removed.

14. Keeping the spreader bar arrangement level, move the turbine casing axially to clear the lifting beam overhang.

15. Lift the spreader bar and turbine casing clear of the unit and set the casing on suitable cribbing to protect the flanges.





NOTES: GUIDE PINS FOUND IN TOOL KIT, MAJOR DISASSEMBLY

GUIDE PINS ARE USED IN BODY ROUND STUD HOLES OF CASINGS & DOWEL HOLES OF BEARING CAPS.

ITEM	GUIDE PIN	WHERE USED
1	248A4880 P4	COMPRESSOR CASING
3	295A9609 P2	TURBINE CASING
4	295A9601 P1	NO. 1 BEARING CAP
5	295A9601 P2	NO. 2 BEARING CAP
6	295A9609 P3	TURBINE CASING
7	295A9609 P1	DISCHARGE CASING
8	295A9600 P1	INLET CASING
9	295A9601 P3	EXHAUST FRAME

Figure HGP-D3. Guide Pin Locations

16. Remove the spreader bar and rigging from the turbine casing.
17. Using overhead crane, re-rig the turbine casing with comealongs or chainfalls to the forward lugs on the casing and wire slings on the aft lugs.
18. Pickup the casing enough to clear the cribbing and by raising on the aft lugs and lowering with the chainfall or comealong on the forward lugs, position the turbine casing on its forward vertical flange for later inspection and or removal of second and third stage nozzle segments.

OPERATION 8 - TAKE TURBINE CLEARANCE CHECKS

1. It is suggested that turbine clearances be taken under the guidance of your GE Field Service Representative. The need to closely monitor the internal gas turbine clearances is critical. Once the turbine casing has been removed, the internal clearances called for on MLI 0404 should be recorded on Inspection Form GT7200. Any clearances not within specified tolerances, as indicated on MLI 0404, should be reported to Engineering prior to removing any parts. Engineering may request that the clearances be verified prior to any further disassembly. The data will be compared to the original baseline clearance data so that a determination for corrective action, if any, can be made. The clearances listed on MLI 0404 should be taken and recorded on Inspection Form GT7200. An example of the MLI 0404 clearance locations are found below.
    a. First stage nozzle support ring to first stage nozzle segment support pad clearance B1, B2.
    b. First stage nozzle clamp clearances (E).
    c. Axial rotor clearances 1F1, 1A1, 2F1, 2A1, 3F1, 3A1.
    d. Axial seal clearances 1F3, 1F5, 2F3, 2A3, 3F3, 3A3.
    e. Axial diaphragm to turbine wheel clearance 1A4, 2F4.
    f. Radial seal clearances 1F2, 1F4, 1A2, 2F2, 2A2, 3F2, 3A2, 3A4.
    g. Bucket tip clearance (1R).
    h. Diaphragm radial seal clearance 1PH, 2PH.
    i. Rotor set points "A", "A" compliment, set point "B", "B" compliments (includes 2PA).
    j. First stage nozzle support ring shim "D".
    k. Shroud to bucket clearances 2SA, 2B, 2S, 3SA, 3B, 3S.
    l. Axial clearance support ring set dimension ("C").

NOTE The use of proper tools for taking clearance measurements is an important factor in obtaining correct readings. Section 2, Standard Practices.

Caution **String tie all hand tools to the wrist to avoid loss in the turbine.**

LEFT SIDE AND RIGHT SIDE (DEFINITION)

1. The left side of the machine by definition is the left hand side when the observer is facing the compressor inlet and looking downstream (direction of air low) toward the generator load coupling end.

ROTOR FLOAT (THRUST CLEARANCE)

1. The maximum total fore and aft movement of the rotor without distorting any of the rotor or bearing mounts is usually 14 to 20 mils. The rotor should be against the active thrust bearing but not loaded to deflect the bearing or casings when the indicator is read.
2. The rotor float is measured by mounting a dial indicator from the No. 1 bearing housing and indicating on the accessory coupling hub. This can be accomplished with the accessory coupling and guard in place. Rotor float is referenced to the No. 1 bearing housing.
3. The rotor is moved from the load gear turbine coupling. Remove the upper half of the load coupling guard. A small hydraulic jack and wood blocking can be placed between the forward load gear bearing housing and the coupling flange to push the rotor forward. The rotor will move "all-at-once" with a solid "bump." Release the jack loading before reading the dial indicator. Reposition the jack and blocking to push the rotor aft from the No. 2 turbine bearing housing. The rotor will "bump," release the jack and read the indicator. Repeat three times to ensure repeatability. Record thrust clearances on Inspection Form GT7200.

Caution **Do not jack against the exhaust plenum.**

TURBINE ROTOR CLEARANCES

1. The rotor must be positioned upstream against the active thrust bearing but not loaded. Any pressure applied in moving the rotor forward for the float check should be released at this time.

2. Prepare for and take clearances around the first stage nozzle.

   a. Apply a downstream load not to exceed 2000 pounds on the first stage nozzle outer sidewall, both sides, to ensure that the nozzle is seated downstream against the first stage shrouds, and the first stage shrouds are seated downstream against the turbine casing.

Note 1: Do not wedge other nozzles simultaneously with first stage nozzle.

Note 2: Unload each stage after clearances are taken on that stage.



Use a hard wood block to transmit load to the nozzle outer sidewall and jack between the nozzle and discharge casing horizontal joint at left and right side.

   b. Take all clearances around the first stage nozzle. Refer to Inspection Form GT7200.

3. Prepare for and take clearances around the second stage nozzle.

   a. With the rotor still in the forward position, position the second stage nozzle and diaphragm (left and right) downstream by first wedging a screwdriver in the first stage shroud to second stage nozzle outer sidewall gap. Ensure that the second stage nozzle aft outer sidewall hooks are seated downstream in the second stage shrouds, and that the second stage shrouds are seated downstream against the turbine shell. See Figure HGP-D4.

   Second, drive a wooden wedge between the forward face of the nozzle diaphragm and the first stage bucket shanks. Drive the wooden wedge until the diaphragm forward male hook is seated downstream against the nozzle female hook. At this point, the wooden wedge should be firmly in place. The downstream seated condition is indicated on Figure HGP-D4 with notation "These gaps must be closed".

   The wooden wedge can be cut from a 2 x 4 and when possible, be made from structural grade of wood. Dimensions of the wedge are shown in Figure HGP-D6. Do not wedge third stage nozzle and diaphragms until all second stage nozzle area clearances have been taken.

   b. Measure the "A" set point and "A" set point complements. See Figure HGP-D7.

      1. Measure the "A" set point, "A", between the forward flat side, not the pointed portion, of the first diaphragm seal tooth and the aft side of the first land of the 1-2 spacer.

2. Measure the "A" set point complements, "AC", between the aft flat side, not the pointed position of each diaphragm seal tooth and the forward side of each spacer land. Proceed with the forward tooth first for "AC1" and proceed downstream for "AC2", "AC3", etc. Record data on Inspection Form GT7200.

c. Measure all radial and axial clearances around the second stage nozzle. Record on Inspection Form GT7200.

Note 1: Do not wedge the third stage nozzles simultaneously with the second stage nozzles.

Note 2: Do not measure the "A" set point or complements with a snap gauge. It is recommended to use a parallel bar and feeler stock. See Figure HGP-D8.

Note 3: The wooden wedge is used only to maintain proper downstream seating of the diaphragm while taking clearances. Do not drive the wedge hard. The wedge should produce inappreciable nozzle elastic deformation. If the wooden wedge has to be driven hard to "shock" the diaphragm downstream, take the wedge out and drive another wedge just until it becomes firmly in place before taking clearances.

4. Prepare for and take clearances around the third stage nozzle.

a. With the rotor remaining in the forward position, position the third stage nozzles and diaphragm downstream similar to the second stage nozzles and diaphragms. See Figure HGP-D5.

Ensure that the third stage nozzle aft outer sidewall hooks are seated downstream in the third stage shroud and the third stage shrouds are seated downstream against the turbine casing. Ensure that the diaphragm forward male hook is seated downstream against the nozzle female hook. This is indicated on Figure HGP-D5, with the notation: "These gaps must be closed."

b. Measure the "B" dimension and "B" dimension complement. See Figure HGP-D9.

1. Measure the "B" dimension, between the forward flat side, not the pointed portion of the second long diaphragm seal tooth and the aft side of the second land of the 2-3 spacer.

2. Measure the "B" dimension complement, "BC", between the aft flat side, not the pointed portion, of each diaphragm seal tooth and the forward side of each spacer land. Proceed with the forward tooth first for "BC1," and proceed downstream for "BC2", "BC3", etc.

c. Measure all other radial and axial clearances around the third stage nozzle. Record

on Inspection Form GT7200. Measure the "B" dimension and "B" complements using similar techniques to "A" set point and "A" complement measurements.

5. Take all other accessible clearances. Record on Inspection Form GT7200.

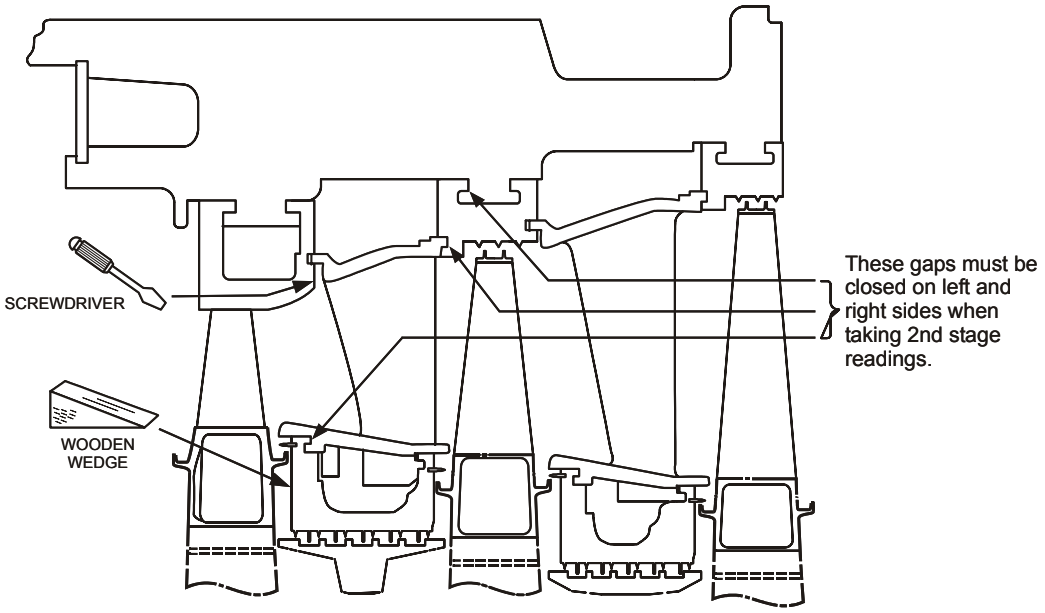


Figure HGP-D4. Loading for 2nd Stage Nozzle Clearances

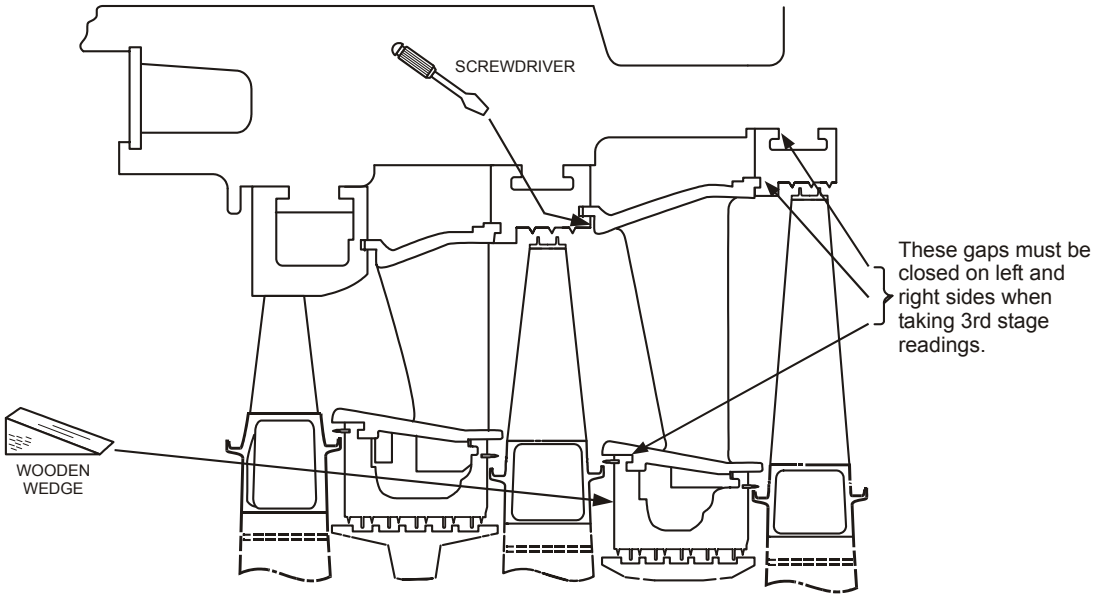


Figure HGP-D5. Loading for Third Stage Nozzle Clearances

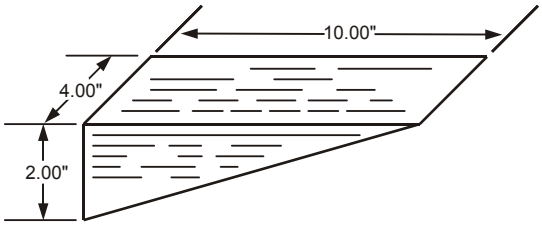


Figure HGP-D6.
Nozzle Clearance Wedging Block

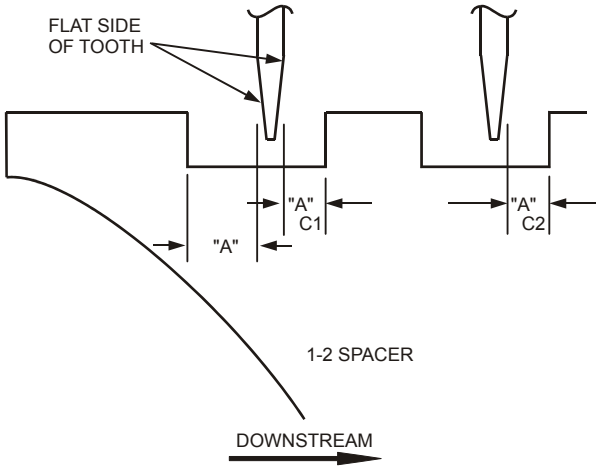


Figure HGP-D7. Setpoint "A"

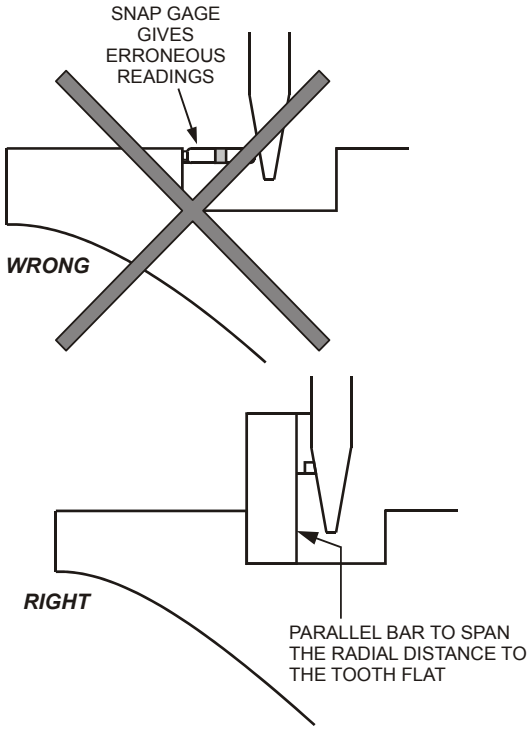


Figure HGP-D8.
Proper Method of Reading Axial Clearances

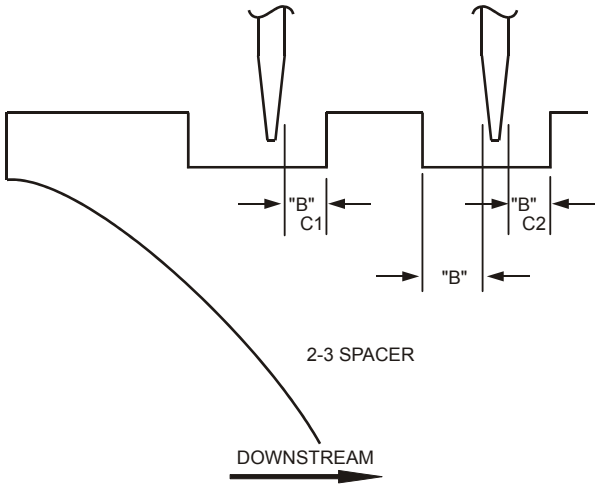


Figure HGP-D9. Set Point "B"

OPERATION 9 - UNBOLT AND REMOVE TRANSITION PIECES

Transition pieces numbers 10, 1, 2, 3, 7, 8 and 9 can be removed radially outward through the turbine casing opening. Transition pieces 4, 5 and 6 must be removed through the corresponding compressor discharge casing opening.

1. Remove the shoulder bolt, lockplate, and aft side seal stop from between 1 and 10 and 10 and 9 transition pieces. Remove the transition piece aft side seals radially outward. Identify the aft side seals and seal stop according to transition piece location.
2. Remove the shoulder bolt(s) and lockplate(s) that secures the number 10 transition piece aft bracket to the first stage nozzle retaining ring. Discard the bolt(s) and lockplate(s). Do not reuse.
3. Move the number 10 transition piece forward and away from the first stage nozzle retaining ring to disengage the floating seals.
4. Slide the transition piece out of the forward support clamp (bullhorn) retaining slots and remove the transition piece radially outward. Identify and tag the transition piece according to combustion casing location.
5. Disassemble the forward mounting bracket lockplates, shoulder bolts, support clamp (bullhorn) and forward mounting block and or shims from the number 10 position. Identify support clamp (bullhorn) block and or shims according to location. Discard the bolts and lockplates. Refer to Standard Practices, Section 2 for proper marking procedure.
6. Repeat Steps 1 through 5 for transition pieces numbers 1, 2, 3, 7, 8 and 9.
7. Remove the shoulder bolt, lockplate, and aft side seal stop from between 4 and 5 transition pieces. Remove the transition piece aft side seals radially outward. Identify the aft side seals and stop according to transition piece location.
8. Disassemble the forward mounting bracket lockplates, shoulder bolts, support clamp (bullhorn) and forward mounting block and or shims from number 4 transition piece. Identify support clamps according to transition piece location. Discard the bolts and lockplates.
9. Remove the aft bracket shoulder bolt(s) and lockplate(s) that secures number 4 transition piece aft bracket to the first stage nozzle retaining ring. Discard the bolt(s) and lockplate(s).
10. Move the number 4 transition piece forward and away from the first stage nozzle retaining ring to disengage the floating seals. Remove the transition piece axially outward through the compressor discharge casing opening.
11. Identify and tag the transition piece according to combustion casing location.
12. Repeat Steps 7 through 11 for transition pieces 5 and 6.

OPERATION 10 - TAKE FIRST STAGE NOZZLE RADIAL CONCENTRICITY CHECKS

Concentricity is the condition wherein the center of one symmetrical feature coincides with the center of another. The measurement of concentricity of the first stage nozzle is the amount of deviation of the nozzle assembly from the center of the hot gas path. Assistance from the GE Field Service Representative is recommended when taking concentricity checks.

1. Prior to taking concentricity checks on the first stage nozzle, ensure that the bottom eccentric pin is installed and that the lower half support ring dowel pins are assembled.
2. The first stage nozzle must be fully assembled prior to taking radial concentricity checks.
3. Take the concentricity readings as shown on Inspection Form GT9250 and Figure HGP-D10. Record all readings on the Inspection Form. Report readings to the GE Field Service Representative for evaluation and disposition.



The 1st stage nozzle is to be concentric with the support ring hot gas path within a tolerance of 0.050 inch, taking into account any ellipticity in the nozzle assembly. See alignment diagram in the Reference Drawing Section of the Service Manual.

OPERATION 11 - REMOVE UPPER HALF FIRST STAGE NOZZLE (WITH SUPPORT PAD AND RETAINER)

There are two types of first-stage nozzle arrangements presently in use, one type uses a support pad seal and a seal retainer for each segment when assembled to the support ring. The second type is known as a universal first-stage nozzle arrangement and has a tangential support pin assembly consisting off an offset bushing, support pin, retainer and lockplate for each segment when assembled to the support ring. Use the disassembly procedure applicable to your unit's configuration. See Figure HGP-D11.

1. Remove the horizontal joint bolting and clamps from first stage nozzle. Do not remove the joint dowels. Identify horizontal clamp for left and right side.
2. Carefully lift the top half of the first stage nozzle using a chain fall. Pay special attention to the support pad seal between nozzle and support ring. The support pad seals must slide out of the inner support ring. See Figure HGP-D11.
3. Clean the support pad seal area, and check for and remove any burrs on edges of support pad seal grooves and tongues.
4. Set the nozzle flat on a piece of plywood or equivalent to protect it from damage.

OPERATION 11 - REMOVE UPPER HALF FIRST STAGE NOZZLE (WITH TANGENTIAL SUPPORT PIN)

1. Remove horizontal joint bolting and clamps from first-stage nozzle.
2. Remove the tangential support pin bolts and retainer plates. Discard bolts and retainer plates.
3. Remove the support pins by breaking the contact between the pin and bushing with a wrench and withdrawing the pins by means of the threaded extraction hole in the center of the support pin.
4. Rig to lift the upper half nozzle using a chainfall attached to the lifting angles on the nozzle. Carefully lift the upper half nozzle clear of the unit and set the nozzle flat on a piece of plywood of equivalent to protect it from damage. See Figure HGP-D11.

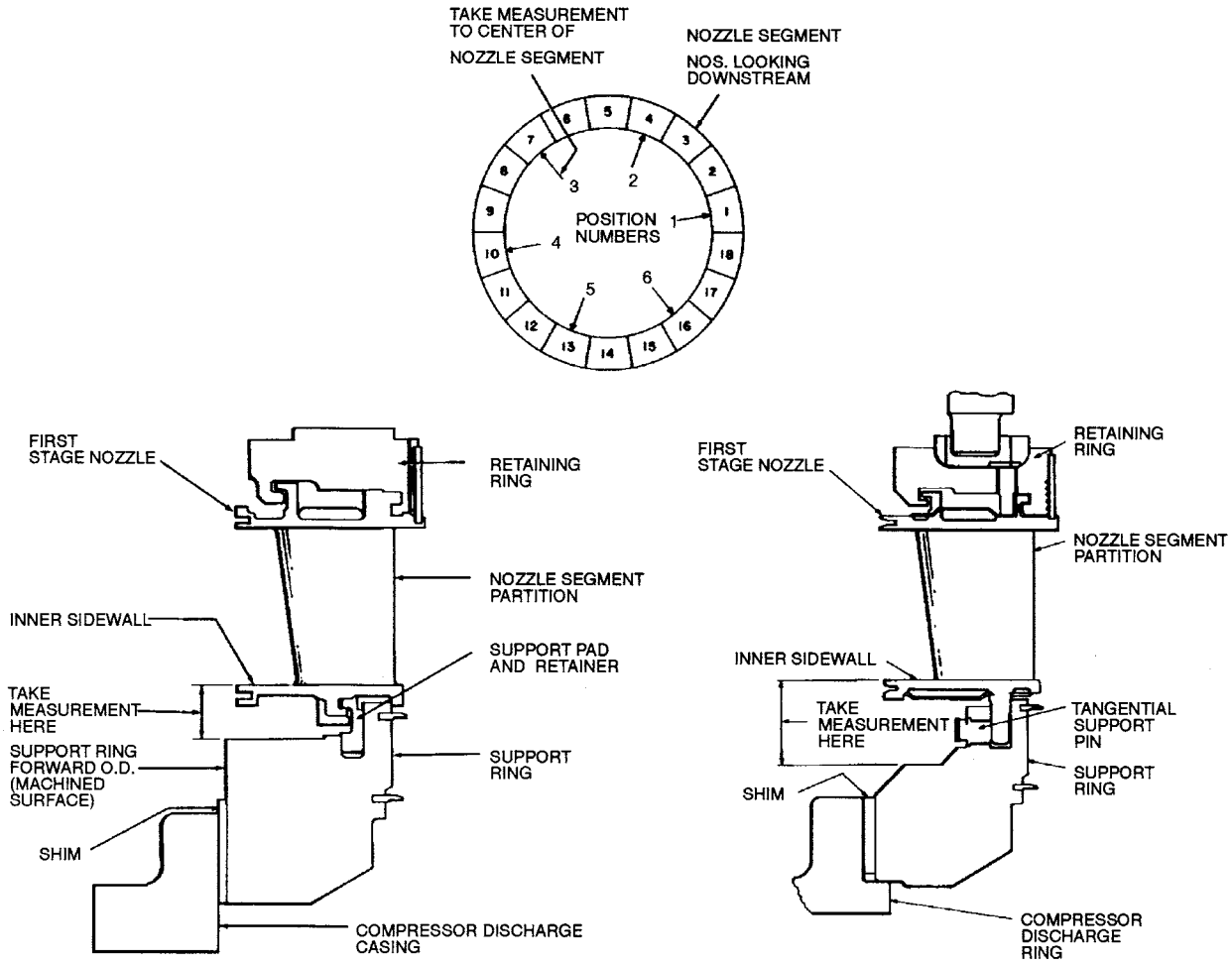


Figure HGP-D10. First Stage Nozzle Concentricity Checks

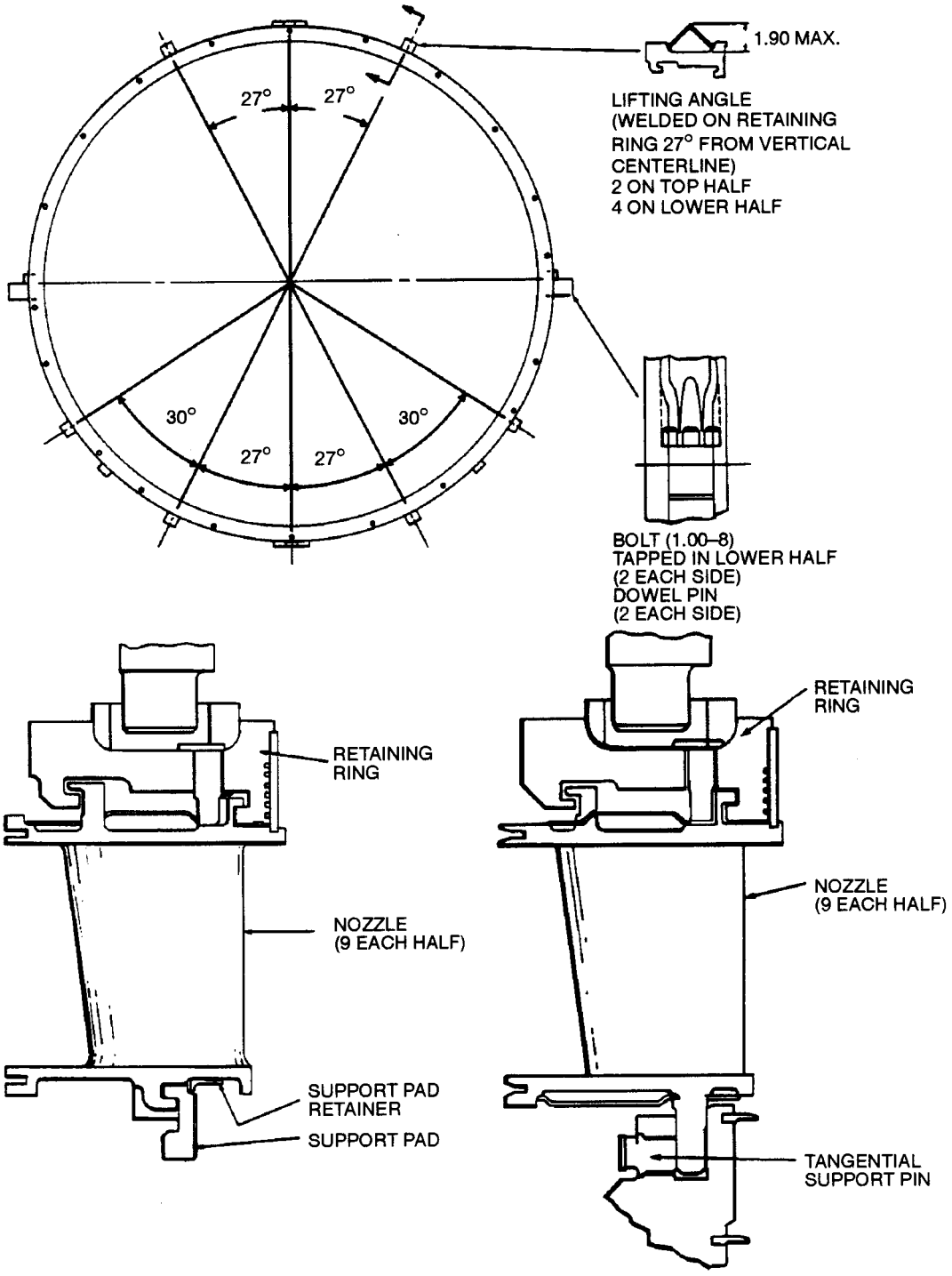


Figure HGP-D11. First Stage Nozzle Lifting Angles for Support Pad and Support Pin Type Nozzle

OPERATION 12 - REMOVE LOWER HALF FIRST STAGE NOZZLE ECCENTRIC PIN

1. Unbolt the eccentric pin retaining strap from the turbine casing and remove the lower half first stage nozzle eccentric pin. Stamp the pin for identification.

OPERATION 13 - REMOVE LOWER HALF FIRST STAGE NOZZLE (WITH SUPPORT PAD AND RETAINERS)

Prior to rolling out the nozzle, remove the horizontal support blocks from the left and right side. Identify the support blocks for left and right side location. These blocks are ground at the factory to obtain proper clearances on the nozzle at assembly.

1. Roll out the lower half nozzle by attaching one end of it to a hook with a straight cable and attaching the other end to a one ton "come-along" or chainfall. By alternately raising on the cable and lowering on the chainfall, the retaining ring and nozzle can be rotated 90 degrees.
2. Reattach the chain hoist to pickup the nozzle at its vertical centerline. Continue the "roll-out" until the lower half nozzle can be lifted clear of the turbine.
3. Set the nozzle flat on a piece of plywood or equivalent and protect same from damage.

OPERATION 13 - REMOVE LOWER HALF FIRST STAGE NOZZLE (WITH TANGENTIAL SUPPORT PIN)

1. Remove the tangential support bolts and retainer plates. Discard bolts and retainer plates.
2. Remove the support pins by breaking the contact between the pin and bushing with a wrench and withdrawing the pins by means of the threaded extraction hole in the center of the support pin.
3. Remove the horizontal support blocks from the left and right side. Identify the support blocks for left and right side location.
4. Roll out the lower half nozzle by attaching one end to a hook with a straight cable and attaching the other end to a one ton comealong or chainfall. By alternately raising on the cable and lowering on the chainfall, the retaining ring and nozzle can be rotated 90 degrees.
5. Reattach the chainfall to pickup the nozzle at its vertical centerline. Continue the roll out process until the nozzle can be lifted clear of the unit.
6. Set the nozzle flat on a piece of plywood or equivalent to protect it from damage.

OPERATION 14 - REMOVE LOWER HALF SECOND AND THIRD STAGE NOZZLE RADIAL RETAINING PINS

The second stage nozzle radial retaining pins are secured to the turbine casing by threaded plugs which are screwed into the casing, each plug is staked in two places. The third stage nozzle radial retaining pins are held in position by plates secured to the turbine casing by hex bolts with lockplates.

1. Remove all the second stage nozzle radial retaining pins, if pins are not stamped as to location, stamp the location upon removal.
2. Remove the holding plates which position the radial third stage nozzle retaining pins. Remove the retaining pins, if pins are not stamped as to location, stamp them upon removal.

OPERATION 15 - REMOVE LOWER HALF SECOND AND THIRD STAGE NOZZLE SEGMENTS (SEE FIGURE HGP-D12)

1. Each nozzle segment can be removed from the turbine casing by rolling it out individually on its outer sidewall forward and aft hook fits. Each diaphragm segment is removed along with its adjacent nozzle segment as one piece. Number each segment per Inspection Form GT9295.



At all points where the segment and pulling cable come in contact there should be ample padding to distribute the load and protect the partition surfaces. Wood, rubber or padded steel plates can be used. Excessive shock and vibration should be avoided. Attach cabling to the center nozzle partition if possible. A maximum force of 2,000 pounds should not be exceeded. The wheelspace thermocouples must be removed prior to removal of the nozzle segments. The "replaceable type" thermocouples in the lower turbine casing are located in segments number 10 and 15 in second stage nozzle, two thermocouples per segment. The third stage nozzle thermocouples, two per segment, are located in segments number 11 and 14.

2. Tooling required for nozzle segment removal are: pulley assembly, one-ton "come-along", and 0.375 inch wire rope or fiber sling. See Special Tools in Standard Practices, Section 2 of the Service Manual.

NOTE Pay particular attention to the nozzle and diaphragm forward, aft and radial seals between each segment when rolling the nozzle segment out. Keep each set of seals with its respective segment. Place the removed nozzle segments on a piece of plywood or equivalent in a protected area. Cover the removed segments to protect them from the environment.

3. Attach cable to nozzle segment 16 and remove the segment using a one-ton come-a-long attached to the overhead crane.
4. Attach cable to nozzle segment 15 and remove.
5. Install pulley assembly in retaining pin hole 16. Pass cable through pulley and attach to nozzle segment 14. Pull nozzle segment to position vacated by nozzle segment 15. Remove pulley assembly and remove nozzle segment 14 as above.
6. Install pulley assembly in retaining pin hole 15. Pass cable through pulley and attach to nozzle segment 13. Pull nozzle segment 13 to position vacated by nozzle segment 14. Move pulley to pin hole 16 and remove nozzle segment 13 as above.
7. Install pulley assembly in retaining pin hole 14. Pass cable through pulley and attach to nozzle segment 12. Pull nozzle segment to position vacated by nozzle segment 13. Move pulley assembly to pin hole 15 and then to pin hole 16 and remove nozzle segment 12 as above.
8. Move to other side of unit and remove segments 9, 10 and 11 in same manner as described above.

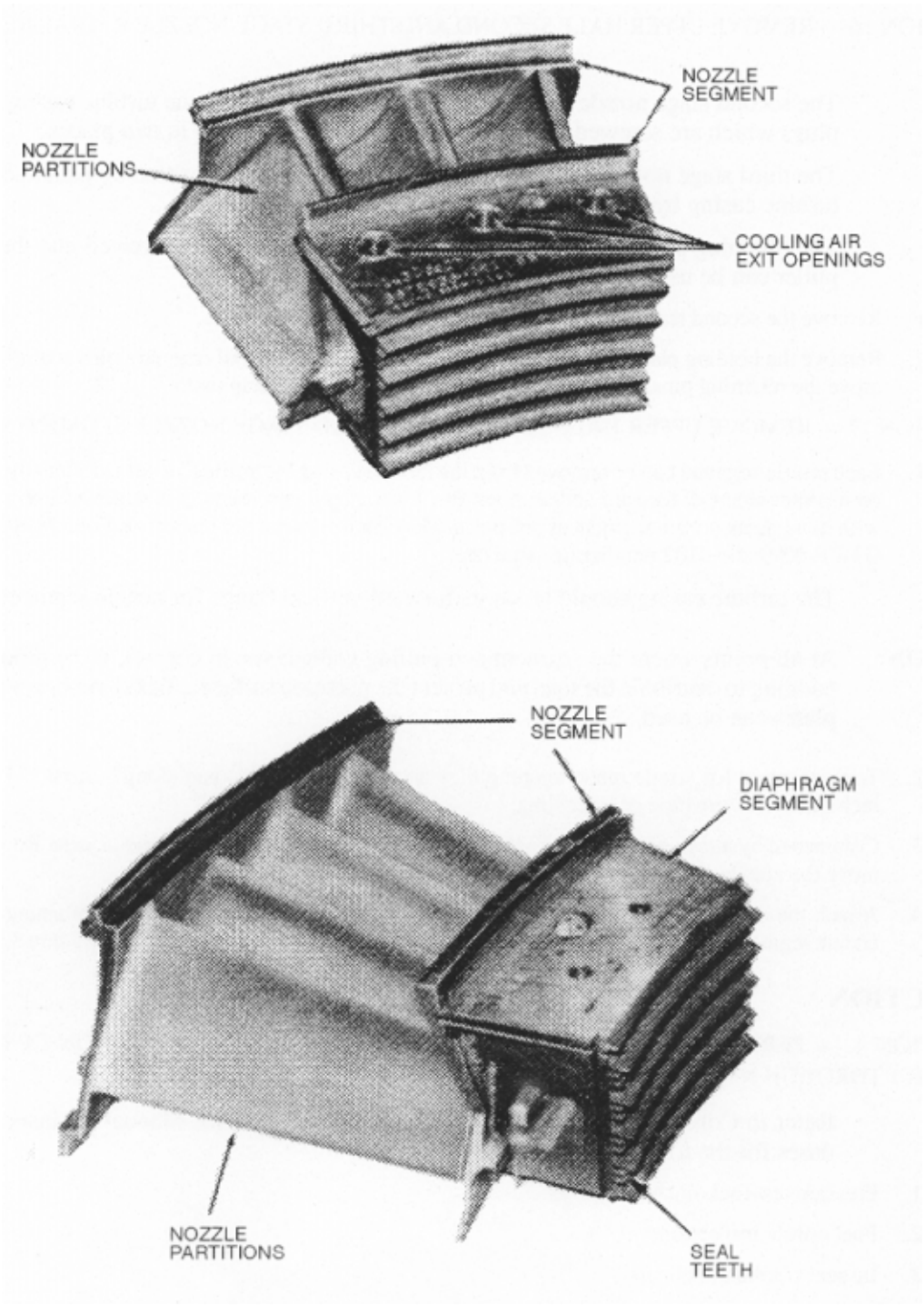


Figure HGP-D12. Second and Third Stage Nozzle Segments

OPERATION 16 - REMOVE UPPER HALF SECOND AND THIRD STAGE NOZZLE RADIAL RETAINING PINS

The second stage nozzle radial retaining pins are secured to the turbine casing by threaded plugs which are screwed into the casing, each plug is staked in two places. The third stage nozzle radial retaining pins are secured in position by plates secured to the turbine casing by hex head bolts with lockplates. The retaining pins are drilled and tapped so that a bolt can be screwed into the head and a puller can be used to extract the retaining pins.

1. Remove the second stage nozzle threaded plugs and the retaining pins.
2. Remove the holding plates which secure the third stage nozzle radial retaining pins and remove the retaining pins. If pins are not stamped for location, stamp them.

OPERATION 17 - REMOVE UPPER HALF SECOND AND THIRD STAGE NOZZLE SEGMENTS

1. Each nozzle segment can be removed from the turbine casing by "rolling" it out individually on its outer sidewall forward and aft hook fits. Each diaphragm segment is removed along with its adjacent nozzle segment as one piece. Mark each segment per Inspection Forms GT9310 and GT9320. The turbine casing should be on its forward vertical flange for nozzle segment removal.

   At all points where the segment and pulling cable come in contact, there should be ample padding to distribute the load and protect the partition surfaces. Wood, rubber or padded steel plates can be used.

2. Tools required for nozzle removal are: pulley assembly, one-ton come-a-long, and 0.375 inch diameter wire rope or fiber sling.
3. Commence by attaching the cable to the nozzle segment closest to the horizontal joint. Remove the nozzle segment using a one-ton "come-a-long."
4. Attach the one-ton come-a-long to a suitable support point. Continue on the remaining nozzle segments and remove in a manner similar to that described in Operation 15, Steps 3 through 8.

INSPECTION

OPERATION 1 - PERFORM COMBUSTION INSPECTION PROCEDURE INSPECTION OPERATIONS NUMBER 1 THROUGH NUMBER 7 AND OPERATIONS NUMBERS 9 AND 10

Refer to Combustion Inspection, Section 4A of the Service Manual, for inspection procedures for the following operations.

1. Pressure test fuel oil check valves.
2. Fuel nozzle inspection.
3. Inspect combustion liners.
4. Inspect crossfire tubes and retainers.
5. Inspect transition pieces.
6. Inspect forward and aft flow sleeves.
7. Inspect combustion outer casings.
8. Inspect and test spark plugs.
9. Inspect and test flame detectors.

OPERATION 2 - INSPECT FIRST, SECOND AND THIRD STAGE NOZZLES

The first stage nozzle may become available for partial or complete inspection at less than hot gas path inspection intervals because of other inspection or maintenance requirements. It is important that these inspection results be accurately interpreted, evaluated, and recorded to minimize operator's maintenance expense while providing high reliability.



Extreme care must be exercised when grit blasting nozzle segments so that no grit enters the nozzle partition cooling holes. Using an air pressure of 40 psig or less, work the blasting nozzle in the direction of normal hot gas flow path across the nozzle partition. Normal hot gas path flow is leading edge to trailing edge. Do not focus on one area as excessive amounts of metal may be removed.

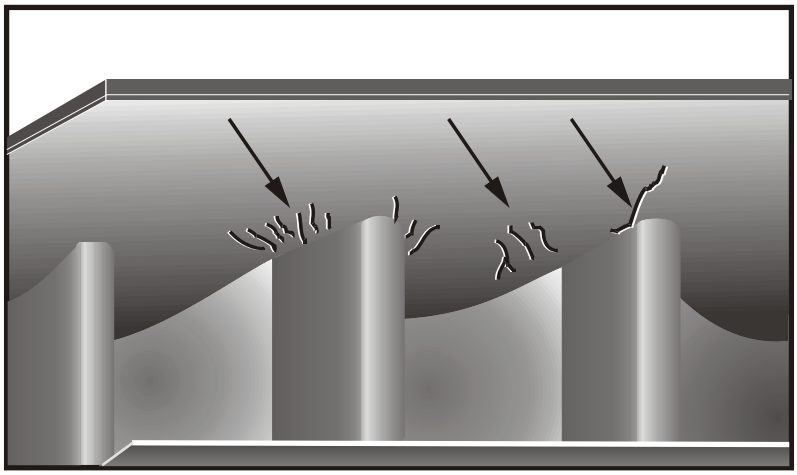
1. Clean nozzles by blasting with aluminum oxide 220 grit or finer. Air blast nozzles to assure removal of all blasting grit from surfaces and openings. Fluorescent penetrant (Zyglo) check the nozzle (including the second and third stage nozzle segments hook fits). Inspect for foreign object damage, erosion, corrosion, and cracks. Refer to Section 2, Standard Practices, Gas Turbine Cleaning of the Service Manual.

Nozzle Cracking

Nozzles experience several thermal gradients during starting as well as high temperatures during loading operation. Such conditions frequently cause nozzle cracking. In fact, cracking is expected. With certain exceptions, this cracking does not impair the efficiency or the mechanical integrity of the nozzle. Figures HGP-I1 through HGP-I5 display typical cracking patterns. The cracks shown are acceptable and do not require immediate repair. If uncertainty arises in regard to any cracks, such as types not represented by Figures HGP-I1 through HGP-I5, contact the GE Field Service Representative for disposition. Obtain specific advice prior to making repairs when nozzle condition is considered beyond established limits.

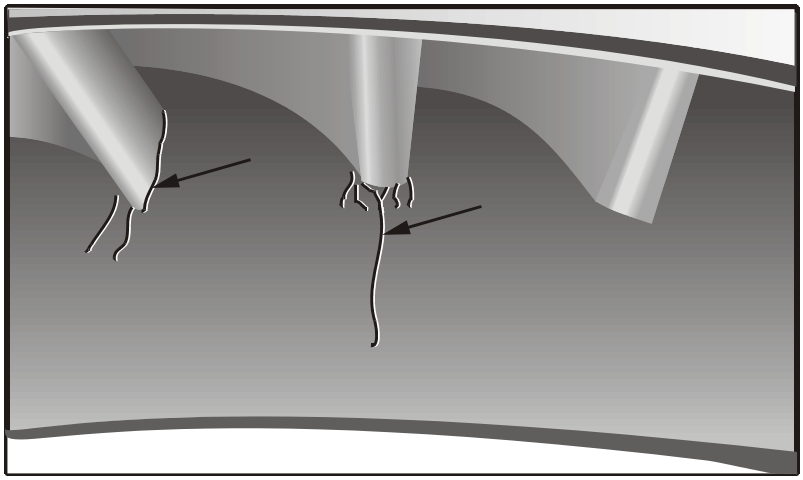
Although to date very few cracks have been found that could impair the mechanical integrity, cracks may still exist. The inspector should be alert for the following which may require immediate disposition (i.e., repair or scrap).

1. **Open cracks in critical areas:** open cracks can permit ingestion of hot gases. An open crack is defined as a crack sufficiently open to pass a 0.005 inch feeler gauge and having a length of 1.0 inch or more. The critical areas are certain regions of the vane and inner sidewalls and are shown in Figure HGP-I6.
2. **Cracks that could cause part breakout:** cracks whose orientation and growth rate are such as to raise questions of the dislodgement of significant portions of the nozzle must be evaluated. See Figures HGP-I7 and HGP-I8 for more details.
3. **Continuous fillet cracks on one vane side:** racks such as this can cause air-foil separation. See Figure HGP-I8 for further details.



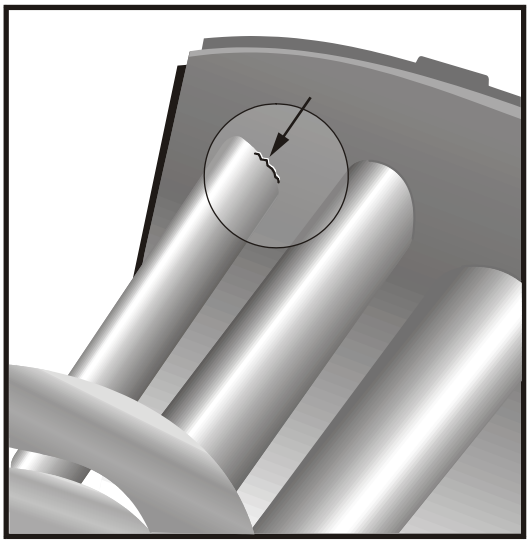
GT71F068

Figure HGP-I1. Nozzle Sidewall Cracks



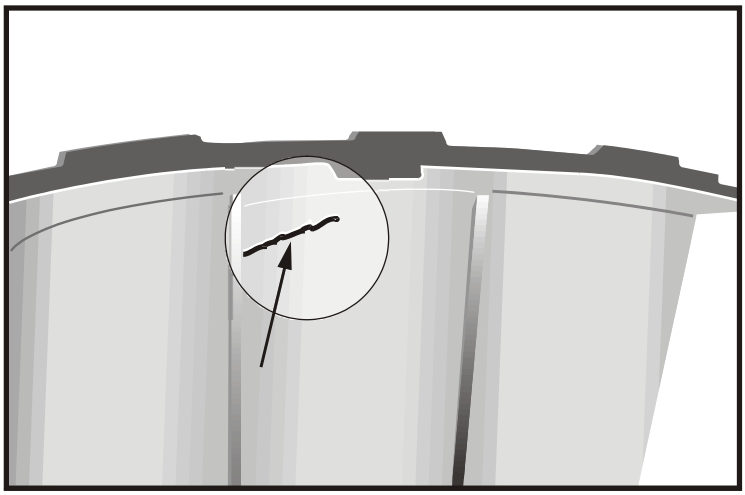
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Figure HGP-I2. Nozzle Sidewall Cracks



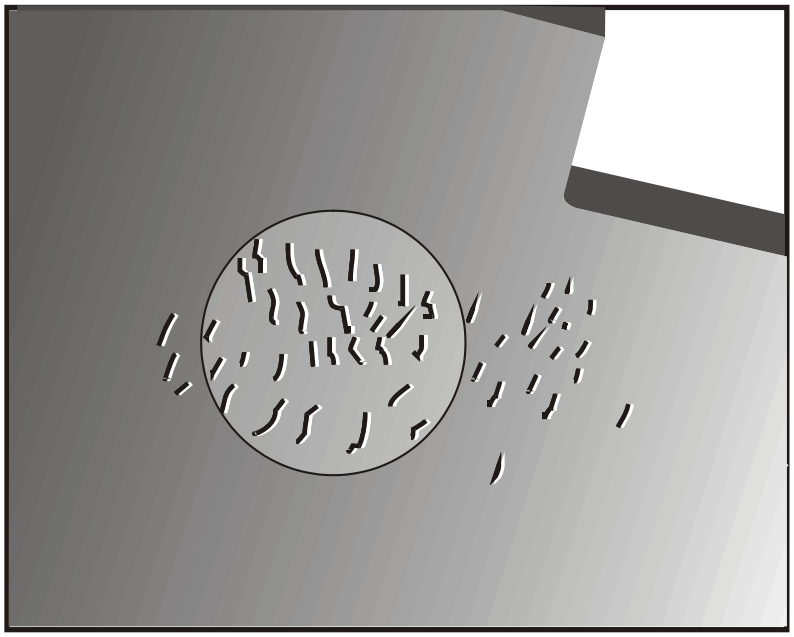
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Figure HGP-I3. Nozzle Partition Fillet Cracks



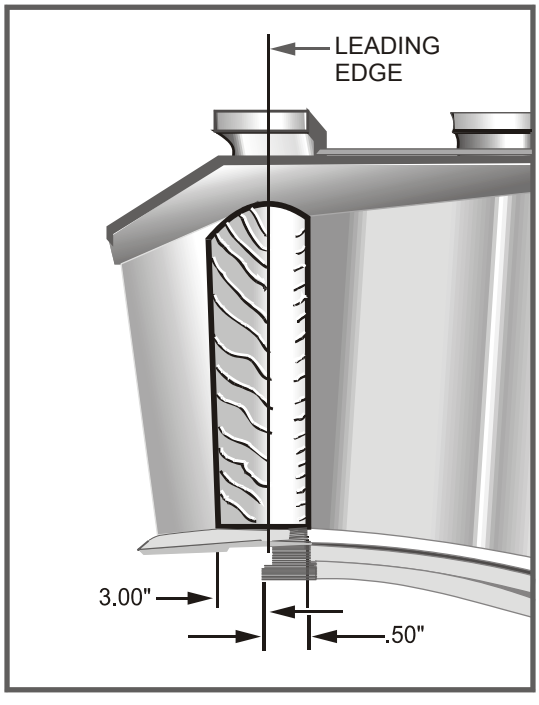
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Figure HGP-I4. Nozzle Partition Fillet Cracks



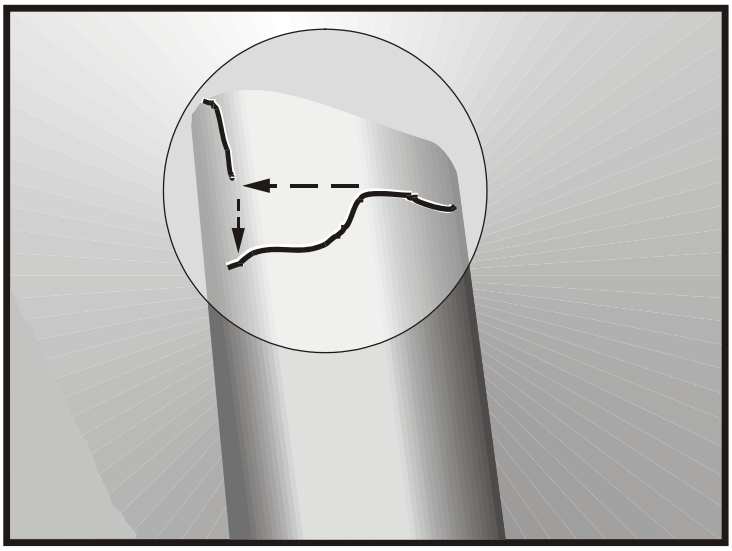
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Figure HGP-I5. Nozzle Sidewall Surface Cracks



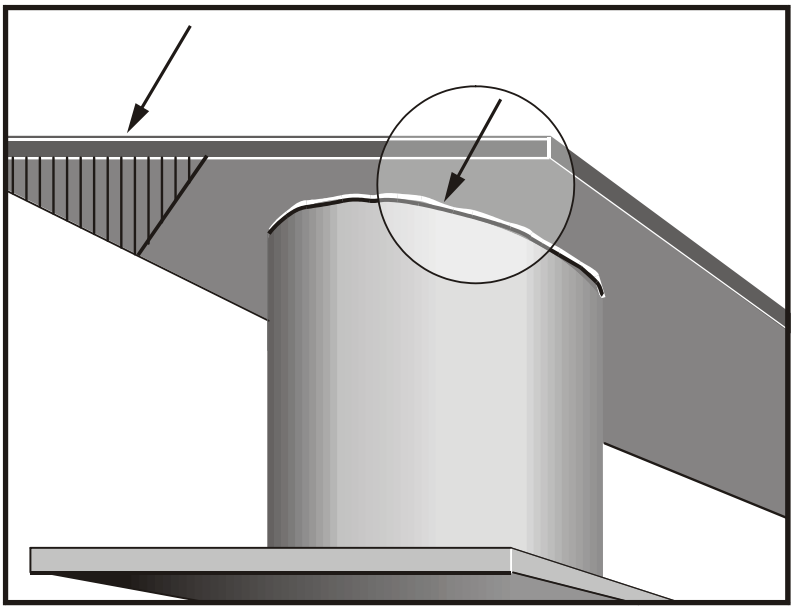
GT71F073

Figure HGP-I6. Unacceptable Cracks



GT71F074

Figure HGP-I7. Unacceptable Crack Propagation



GT71F075

Figure HGP-I8. Unacceptable Crack Propagation

Nozzle Damage from Foreign Objects

Nozzles can be severely damaged by foreign objects in the gas stream. This damage can impair the structural integrity and aerodynamic performance of the nozzle. It should be evaluated using the following criteria.

1. All tears and breaks must be evaluated by the same criteria given above for cracks.
2. Raised metal must be blended to the adjacent surface contours.
3. Nicks and chips must not exceed 50% of the wall thickness.
4. Cooling holes must be open enough to pass a 0.050 inch wire.

Corrosion or Erosion of Nozzles

There have been no cases of severe corrosion or erosion of nozzles to date. The following guidelines are provided in the event that such cases develop.

1. The allowable amount of vane region metal removal is 50% of wall thickness (0.100 inch).
2. Sufficient trailing edge material must be maintained to prevent nozzle cooling hole exposure.
3. Area and harmonic checks must be performed where significant material has been removed.

Recommendations

a. First stage nozzles should not be removed for repair except at normal hot gas path inspection intervals unless the above guidelines have been exceeded.

b. If the nozzle is removed earlier than hot gas path inspection interval for other reasons, all cracks, damage, etc. should be carefully mapped using proper non-destructive test procedures and reported to the GE Field Service Representative. Do not send to repair facility until mapping has been reviewed and specific Gas Turbine Division recommendations are received from the GE Field Service Representative.

c. At all other times that the nozzle becomes partially or wholly accessible, the nozzle should be inspected as completely as possible using appropriate techniques and results reported to the GE Field Service Representative.

d. All inspections should be reported using the GT series Inspection Forms. It should be noted that extreme care should be used to ensure accuracy. False indications can occur when accessibility is limited. Such false indications could result in major expense and lost time if not scrutinized and properly interpreted. Record condition of nozzles on Inspection Forms GT9255, GT9260, GT9270, GT9295, GT9310, and GT9320.

OPERATION 3 - INSPECT FIRST, SECOND AND THIRD STAGE TURBINE BUCKETS


Caution

Turbine buckets should be carefully examined each time the upper half turbine casing is removed. Such examination can reduce the risk of major damage from the failure of a previously damaged bucket. At the same time, judgment is necessary to avoid replacement of adequate buckets. The recommendations contained herein are intended to help in making this judgment, and the criteria for continued service or repair of buckets are furnished as a service to the customer. There is variance in customer practices relating to operation and maintenance, with accompanying variation in influence upon satisfactory operation of gas turbine components. GE has little knowledge or control over these practices. Accordingly, the responsibility for decisions as to continued usage or replacement of buckets must remain solely with each customer.

1. The results of a bucket inspection should be documented along with the relative information on unit operation and fuels. The inspection reporting forms, should be used to record and communicate to GE the exact location and type of abnormality observed. These bucket inspection records will be filed for each particular turbine and used as a basis for present and future recommendations. Use Inspection Form GT7000, GT9015, GT9055, and GT9075.

   All records and questions arising on the subject should be communicated to the GE Field Service Representative.

   Bucket deterioration may be classified as cracks, dents, missing metal, and corrosion. Visually inspect the turbine buckets for these abnormalities. See Figure HGP-I9 for first stage bucket details, and Figure HGP-I10 for second and third stage bucket details.

Note 1: It is recommended that bucket rework be done by qualified GE personnel.

Note 2: In all cases, the bucket rework area must be cleaned and fluorescent penetrant inspected with ZL-22A, or equivalent, when rework is completed.


Caution

Do not attempt to clean buckets prior to inspection.

2. Visually inspect the vane section, platform and dovetail areas, using a ten power (10X) glass examine the critical area of vane section (root fillets, top 25% of tip including squealer tips and shrouds, and trailing edges). Any suspected cracks should be locally cleaned using aluminum oxide 220 or 240 grit paper and hand clean to prepare the surface area for fluorescent penetrant testing. Surface temperatures during polishing operations should not exceed 100F or ambient temperature, whichever is greater. High temperature will cause metal to yield and flow, covering defects so that they cannot be detected.



All first stage buckets have a corrosion resistant protective coating. This coating is only 0.002 inch thick and extreme care must be taken to avoid damaging this coating when cleaning, repairing or handling these buckets. Coated buckets must be cleaned with aluminum oxide 240 grit at an operating pressure of not more than 40 psi. Reworking of coated buckets is allowed, but the rework should be held to a minimum so that no more coating is removed than is absolutely necessary to effect the repair.

First Stage Buckets

a. **Cracks** - Cracks on first stage buckets are unacceptable with one exception: radial cracks on the bucket squealer tip less than 0.2 inch long are permissible. These cracks usually occur as a result of Foreign Object Damage (FOD). Crack detection requires the use of fluorescent penetrant on buckets. Zyglo ZL-22A penetrant is desirable because it will detect tight cracks which red dye type penetrant cannot detect. Radial cracks in certain areas of the squealer tip may be remedied by removing material. The limits are covered the Missing Metal section.

b. **Dents** - Dents are permissible with the following limits.

   1. The dent must not contain any cracks per the above paragraph.

   2. The dent must not obstruct the diameter of, or penetrate into, a cooling air hole. The cooling hole closest to the leading edge should pass a 0.067 inch diameter wire or next larger standard ball to be acceptable. Holes 2 through 12 must pass a 0.050 inch diameter wire or a next larger standard ball to be acceptable. The trailing edge hole must pass a 0.034 inch diameter wire or next larger standard ball to be acceptable.

   3. A dent must not be more than 0.090 inch deep. Therefore in areas where metal thickness to a cooling hole is not the limiting factor, a dent may be blended to remove resulting cracks or metal displacement, providing the 0. 090 inch limit is not exceeded.

c. **Missing Metal** - Missing metal on first stage buckets occurring in the squealer portion can be repaired by blending. Missing metal or radial cracks in the squealer portions can be corrected within the following limits:

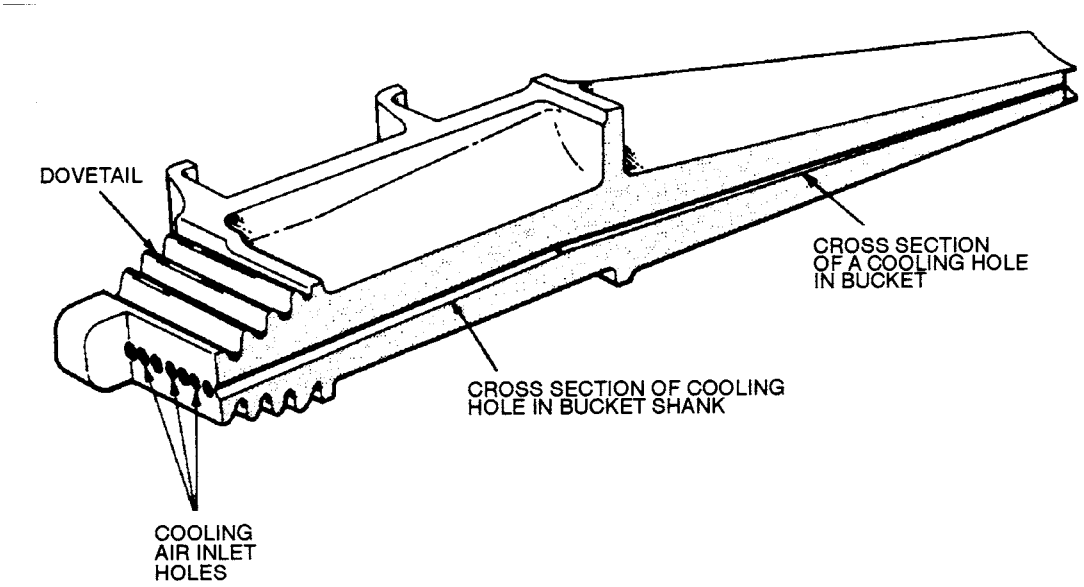
1. The squealer tip wall may be removed for a surface distance of 1.0 inches from the leading edge on the suction side of the bucket. The squealer tip wall may be removed for a surface distance of 0.50 inch from the leading edge on the pressure side of the bucket. See Figure HGP-I11, Squealer Tip Repair Limit.

2. Missing metal, dents and blending cracks in the leading edge must not extend more than 0.750 inch down from the top edge of the bucket. No more than a total of five buckets may have leading edge blending below the squealer section without a resultant loss in performance. See Figure HGP-I11, Leading Edge Repair Limits.

d. **Corrosion** - First stage buckets are more susceptible to damage by corrosion, erosion, and oxidation than second or third stage buckets. This condition usually first appears on the leading edge of the airfoil as a flattening and roughening of the surface; however, the entire airfoil is susceptible to corrosion, erosion or oxidation attack.

Direct measurement of this wearing is difficult, if not impossible. However, general criteria can be established. The cooling hole closest to the leading edge is approximately 0.15 inch from the leading edge, and about 0.06 inch from the airfoil suction side surface. The corrosion pattern on the leading edge will allow a maximum of 0.088 inch corrosion before a breakdown of the hole closest to the leading edge occurs. This breakdown appears as a depression on the suction side surface, near the leading edge. If this condition appears on the first stage buckets, replacement is required.

If some lesser amount of corrosion is observed, judgment is required to weigh the possibility of a corrosion related failure against the cost of replacement of buckets at the time of inspection. Realize that 0.088 inch of corrosion constitutes a total corrosion life. The fraction of 0.088 inch removed is the fraction of the life used. (It is assumed that corrosion progresses linearly with time.) Replace the buckets if total corrosion life is predicted before the next planned inspection.



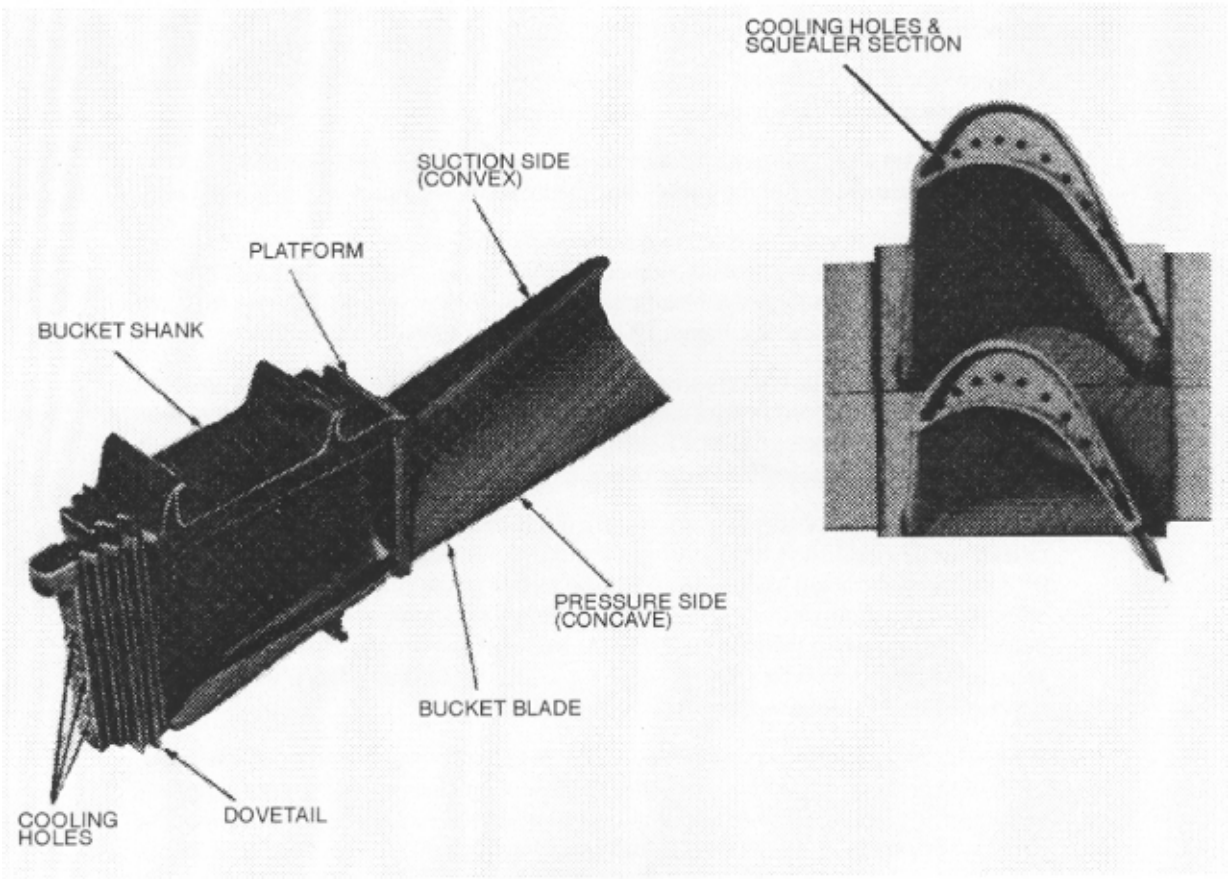


Figure HGP-I9. First Stage Bucket Details

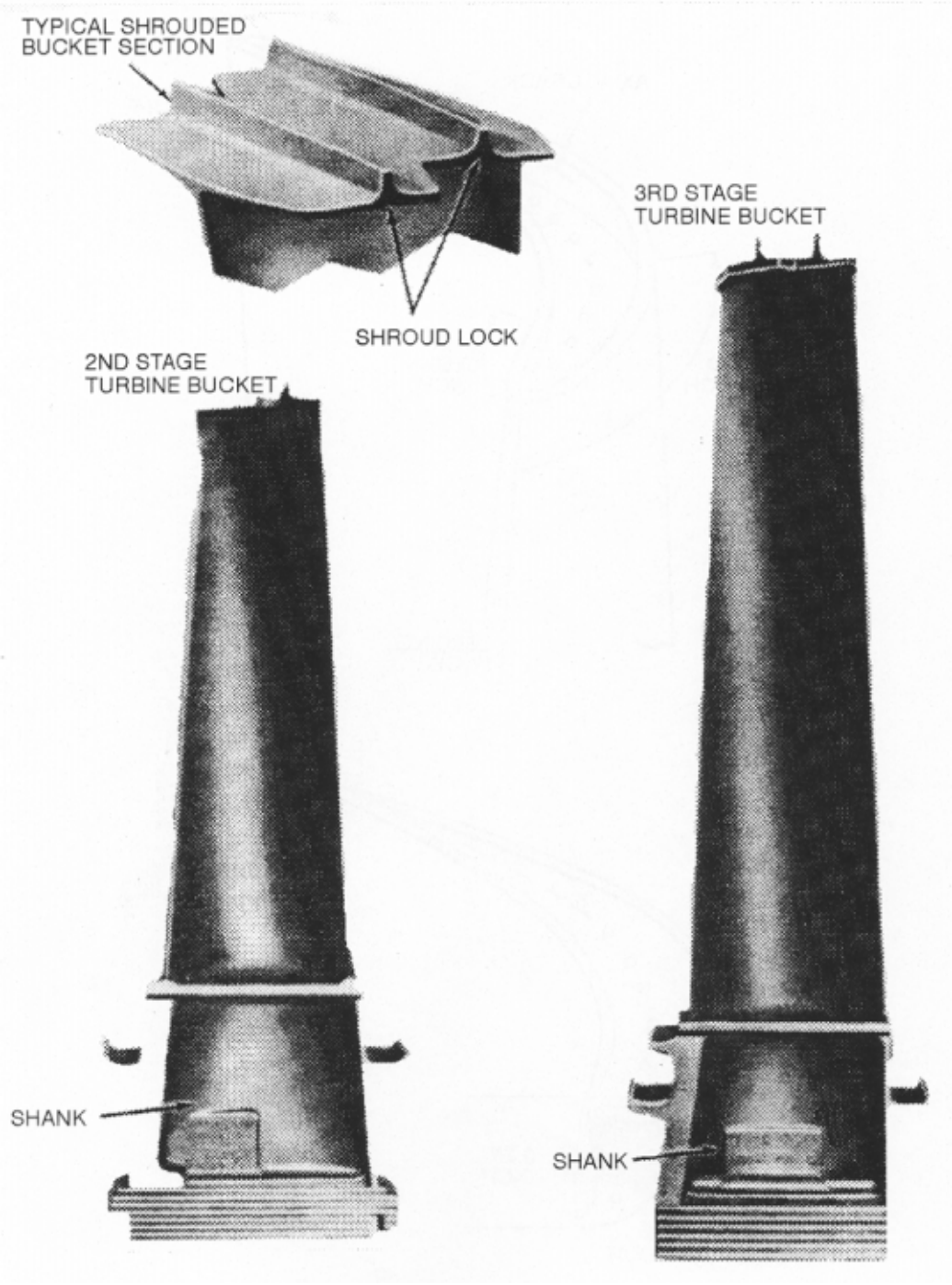


Figure HGP-I10. Second and Third Stage Bucket Details

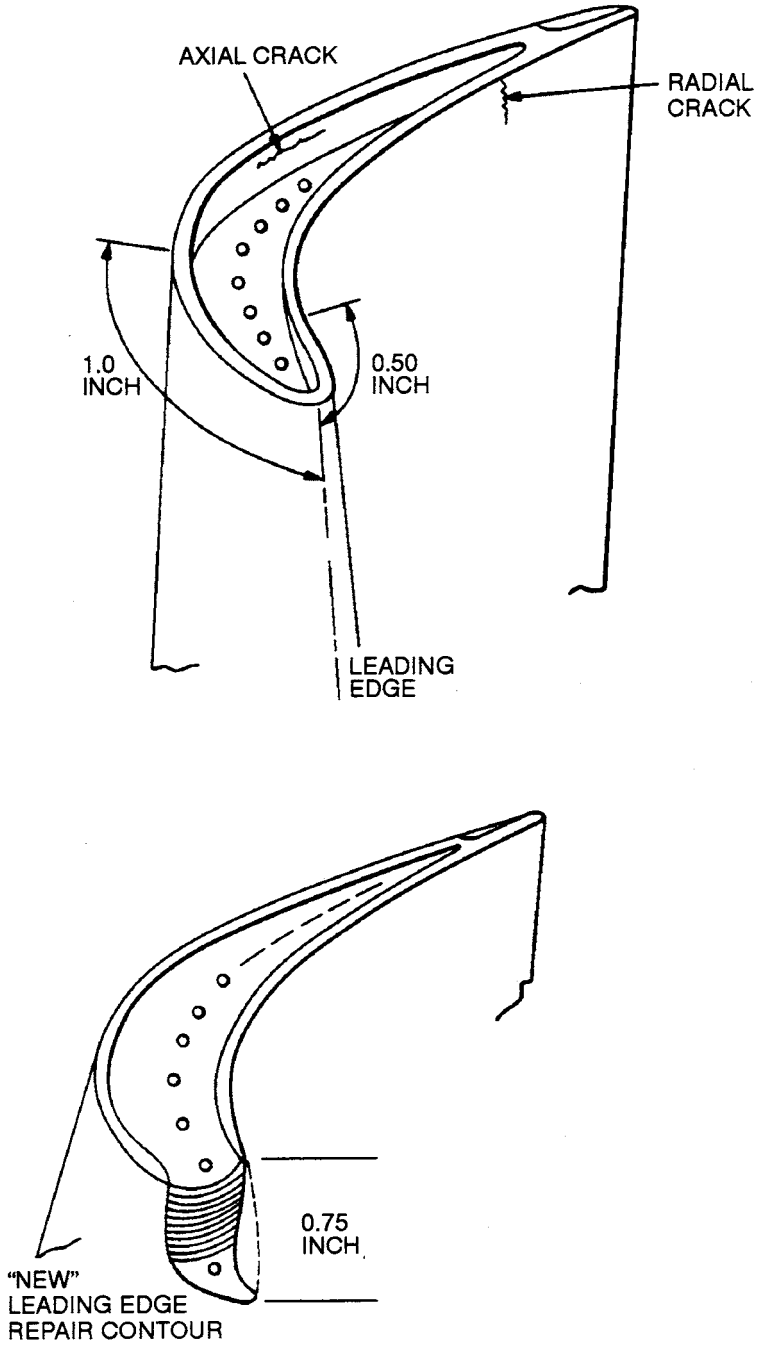


Figure HGP-I11. First Stage Bucket Repair Limits

Second and Third Stage Buckets

a. **Cracks** - Cracks also occur in the second and third stage buckets due to foreign object damage. These cracks usually occur at the airfoil leading edge in the outer span sections.

1. Second stage bucket cracks are repairable if located in the outer 20% of the airfoil and up to 0.250 inch in length. See Figure HGP-I12, Second Stage Bucket Repair Limits.

2. Third Stage bucket cracks are repairable if located in the outer (1/3) one-third of the airfoil and up to 0.350 inch in length. See Figure HGP-I12 Third Stage Bucket Repair Limits. These cracks can be repaired by blending the airfoil in the area of the crack while maintaining radii at all intersection points. The leading edge radius must also be retained in the repaired area.

b. **Missing Metal** - Missing material in the second and third stage buckets is acceptable if the airfoil can be contoured to the same limits as crack repair.

c. **Dents** - Dents are acceptable, as long as the dent is not located in the bucket root fillet (interface between airfoil and platform) or in the shroud fillet (interface between airfoil and shroud).

Dents containing cracks or those located in the shroud fillet are repairable. Repairable dents should be hand polished to remove foreign material in the area of the dent, and fluorescent penetrant inspected for cracks. Blend smooth all dents.

d. Air-cooled second stage buckets must pass a 0.050 inch diameter wire through all four holes from the bucket tip to beyond the platform. An additional check should be made to insure that the passage ways through the bucket shank is open and clear.

e. **Corrosion** - Corrosion on second and third stage buckets to a depth of more than 0.025 inches has not been observed. Such a condition is acceptable. If a more serious corrosion is noted, the observations should be reported to the GE Field Service Representative.

Radial Seal Pin Grooves

1. Cracks and damaged material may be blended within the following allowances.

   a. Radial Seal Pin Groove Repair Limits. See Figure HGP-I13.

      1. Region I - no missing material allowed.

      2. Region II - uter lands 0.500 inch missing metal on both sides.

      3. Region III - no missing material allowed.

      4. Region IV - blending allowed in this region.

Wheelspace Seals/Angel Wings

1. Rubbed wheelspace seals are suitable for continued operation as long as the corresponding wheelspace temperatures are within operating limits.
2. Axial cracks are common and are acceptable in region "A" providing they do not penetrate the seal to shank radius "B" shown in Figure HGP-I14.
3. Seals with the upturned edge rubbed off are unsuitable for use. See Figure HGP-I14.

Shroud Tip Rework

1. Rework and metal removal is allowed in the cross-hatched areas. No rework is permitted which penetrates below the center section. See Figure HGP-I15.



Equal amounts of material should be removed from the leading and trailing edges.

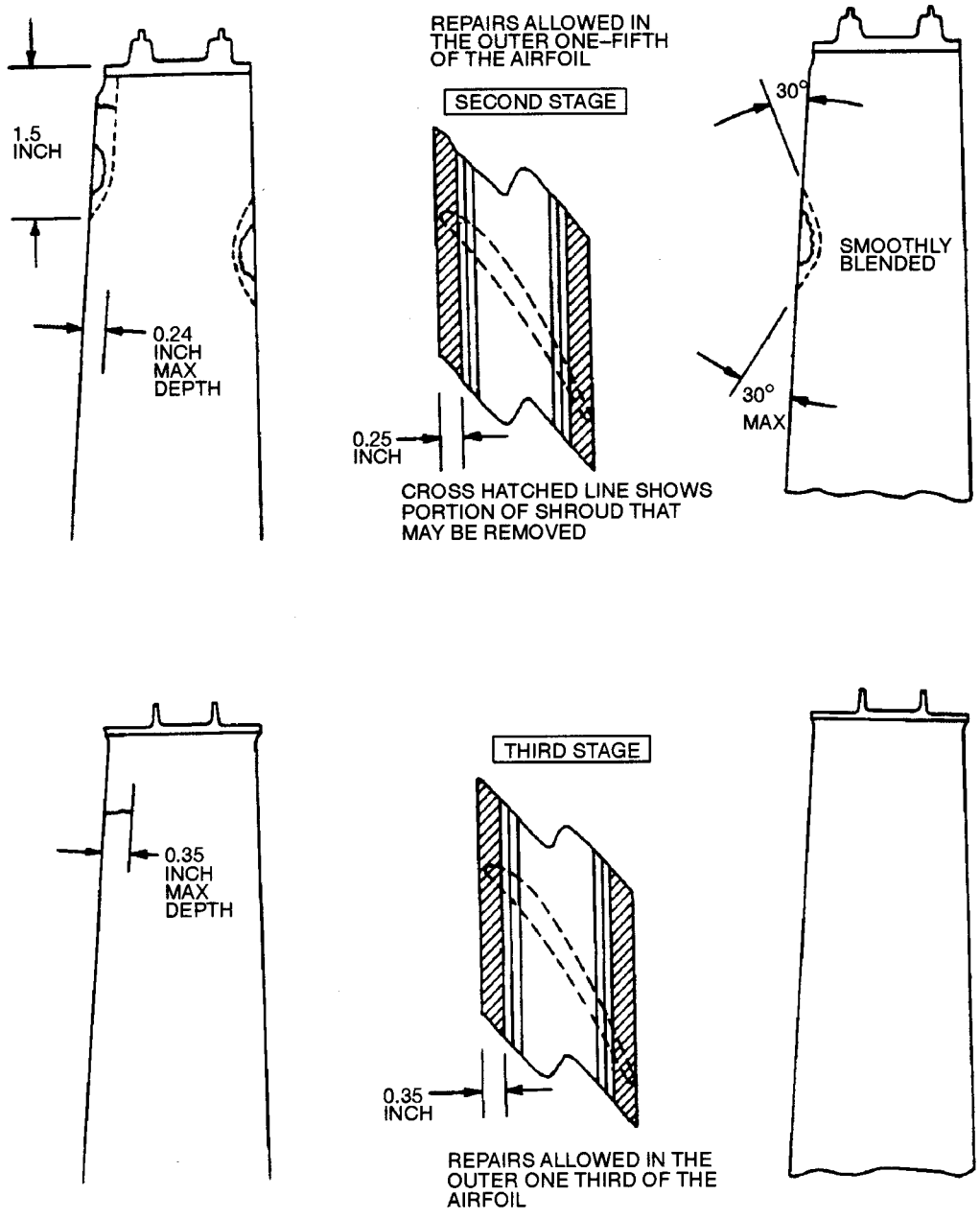


Figure HGP-I12. Second and Third Stage Bucket Repair Limits

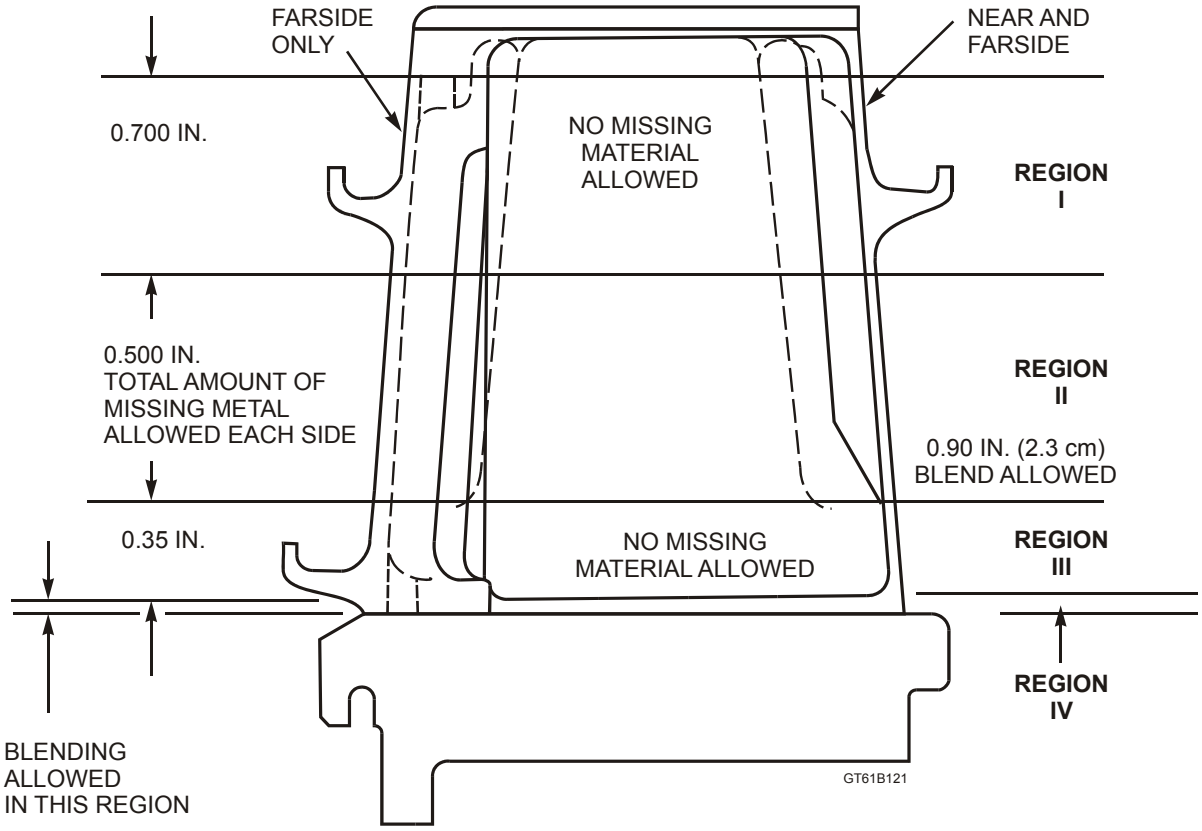


Figure HGP-I13. Radial Seal Pin Groove Repair Limits

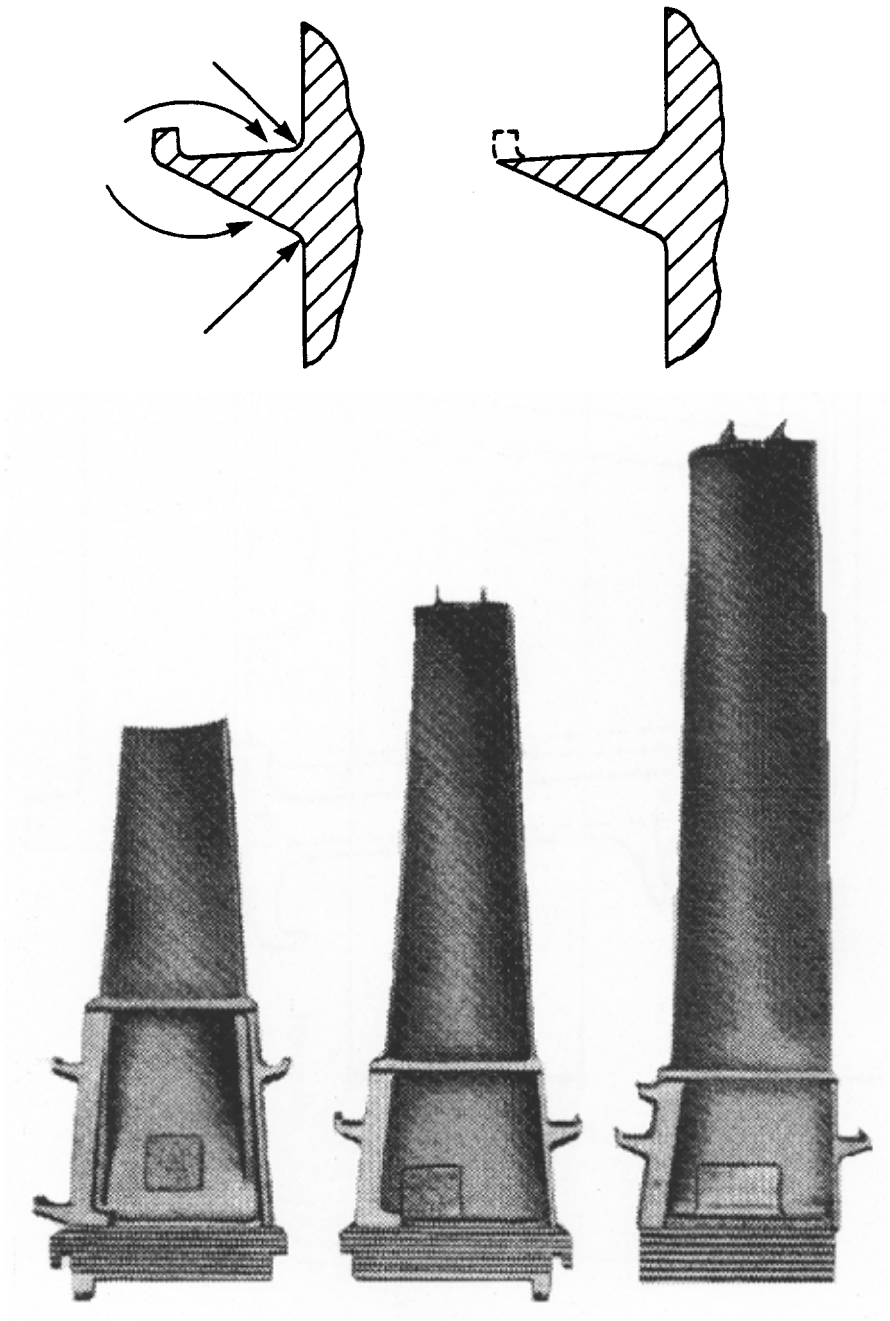


Figure HGP-I14. Wheelspace Seal Angel Wings Cross Sections

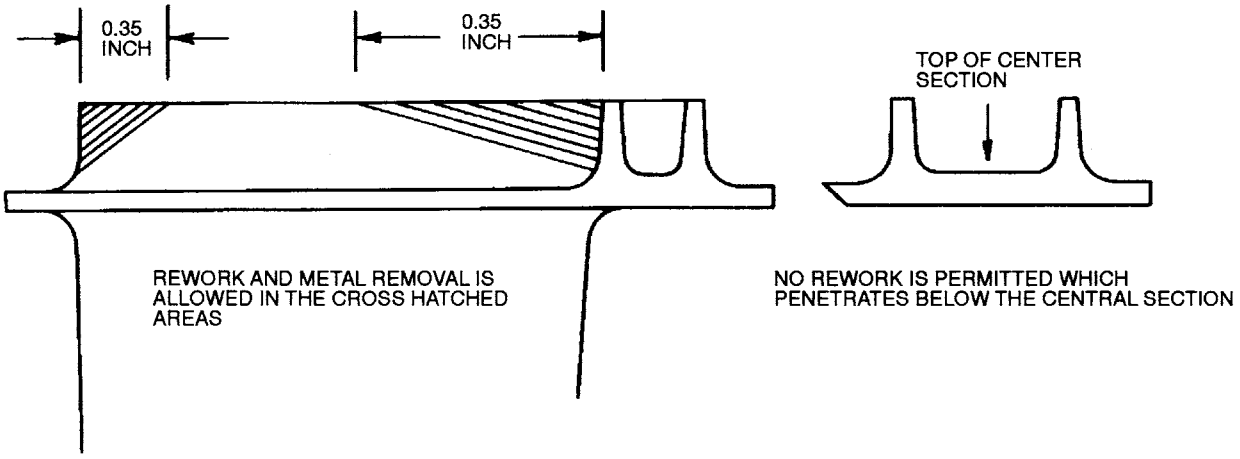


Figure HGP-I15. Shroud Tip Rework Limits

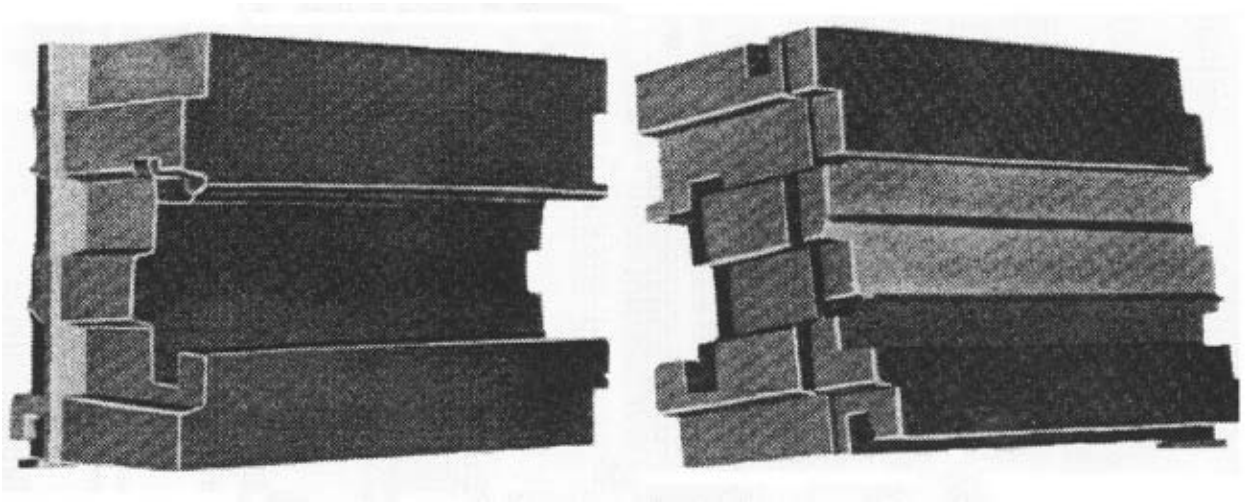


Figure HGP-I16. Turbine Shroud Blocks

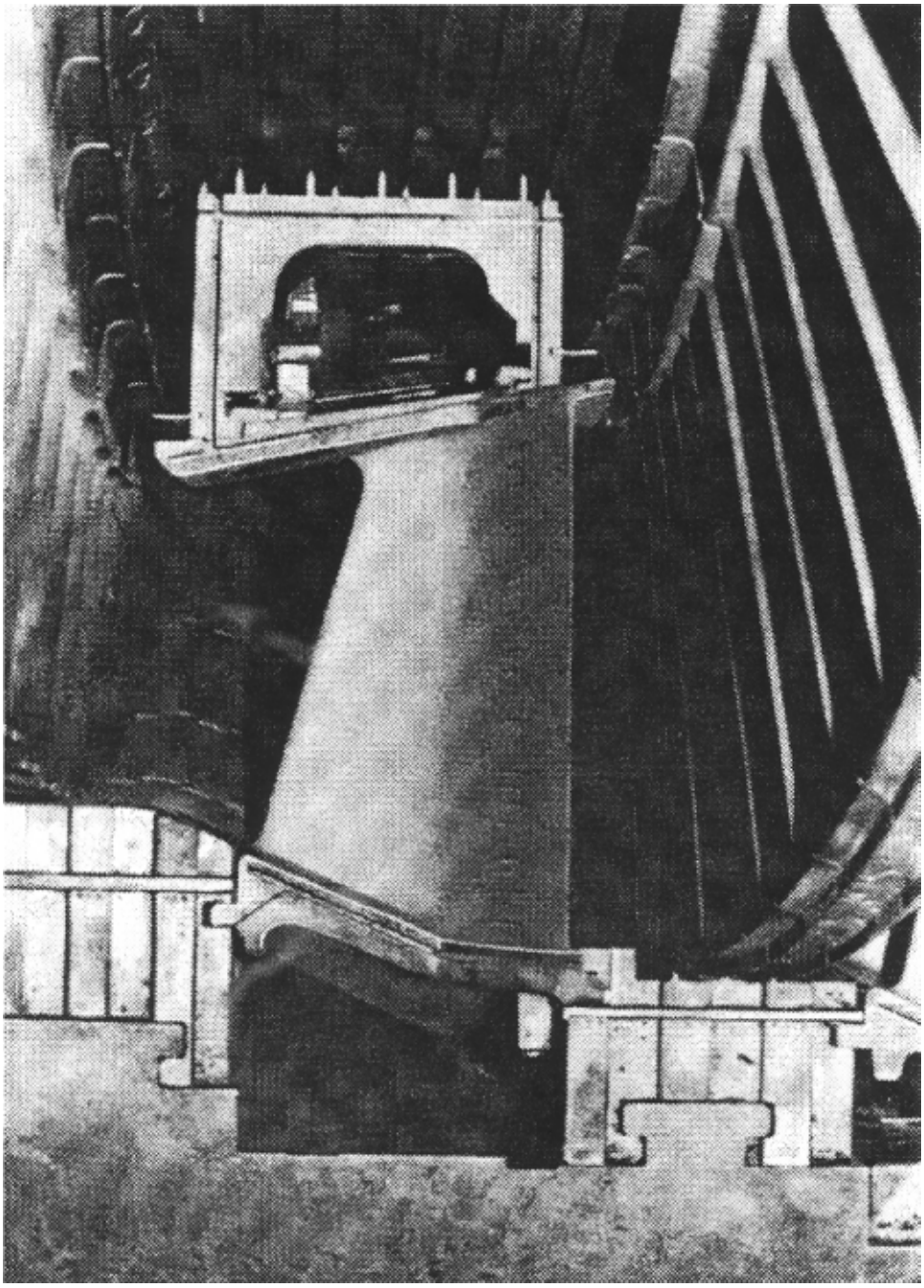


Figure HGP-I17. Second Stage Nozzle and Diaphragm

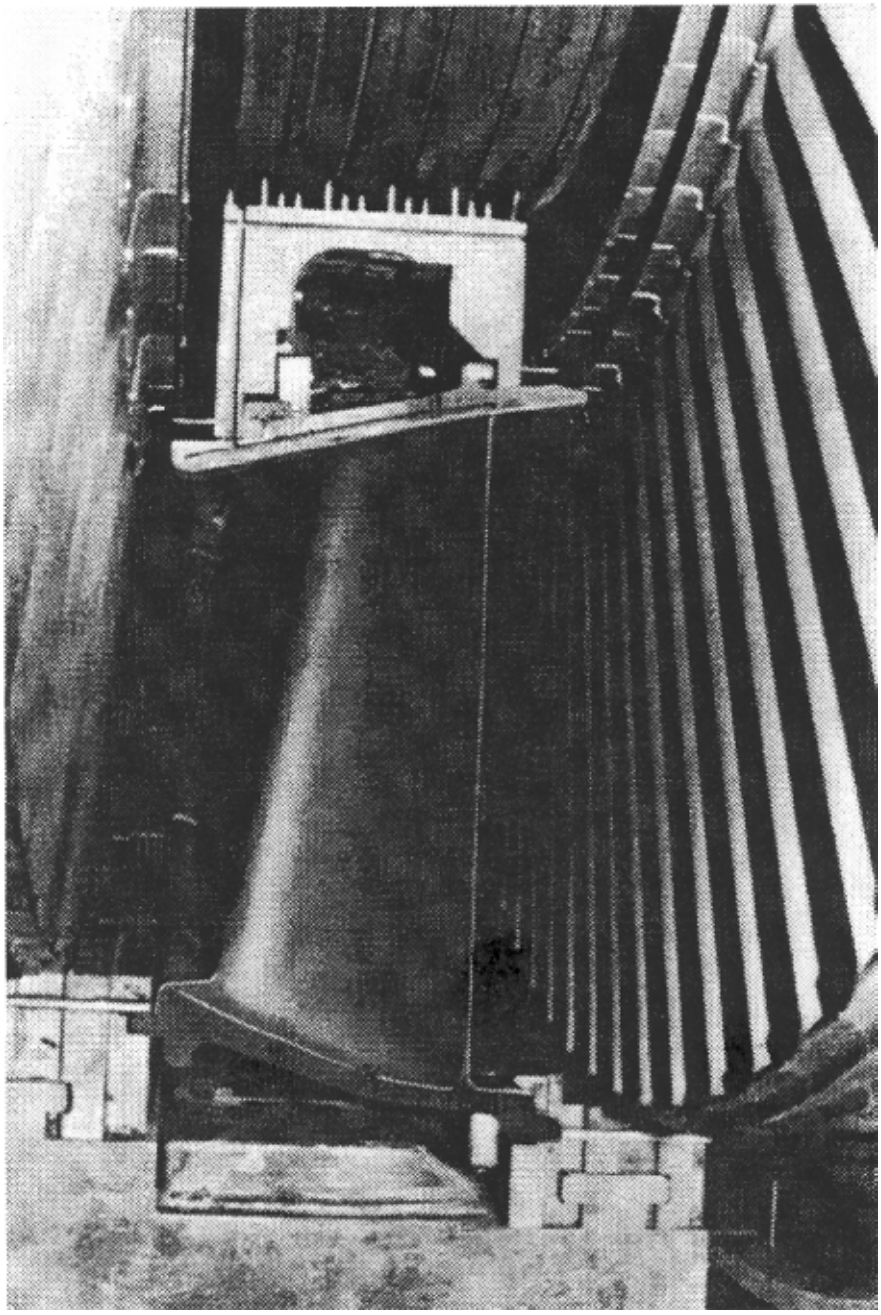


Figure HGP-I18. Third Stage Nozzle and Diaphragm

OPERATION 4 - INSPECT SHROUD BLOCKS

1. Visually check the shrouds segments and hook fits for deposits, corrosion, erosion (pitting), cracks and nicks or dents. Record findings on Inspection Form GT9095, GT9115, and GT9145.
2. Visually inspect the shroud seals for distortion, rubs, wear, and missing pieces. See Figure HGP-I16. The shrouds do not have to be disassembled from the turbine casing for this inspection.

OPERATION 5 - INSPECT SECOND AND THIRD STAGE NOZZLE DIAPHRAGMS

1. Clean nozzle assembly by blasting with aluminum oxide 220 grit or finer as detailed in Section 2, Standard Practices, Gas Turbine Cleaning in the Service Manual.
2. Check for cracks using fluorescent penetrant. Zyglo ZL22A is preferred. Record findings on Inspection Form GT9165 and GT9180.
3. Make a visual inspection for foreign object damage, burning, corrosion, erosion, and excessive deposit buildup. Report findings on Inspection Form GT9165 and GT9180.

   NOTE Each diaphragm segment is normally removed along with the nozzle segment. The diaphragm segments are not normally separated in the field to make checks.

4. Make a visual inspection of all diaphragm seal teeth, first-stage aft wheelspace seals, second-stage forward wheelspace seals and second-stage aft and third-stage forward wheelspace seals. See Figures HGP-I17 and HGP-I18.
5. Note the condition on Inspection Form GT9165, GT9180 and GT9350.
6. Inspect diaphragm teeth as follows. Observe the worst rubbed tooth and measure its tooth height. Compare this measurement to the height of the tooth on the diaphragm segment previously positioned at the horizontal joint w here the tolerance of the gap 1PL and 1PH or 2PL and 2PH (GT7200) is known. Then see if the gaps are within the minimum specified allowance for the worst diaphragm tooth condition.

OPERATION 6 - MAKE FIRST STAGE NOZZLE ELLIPTICITY CHECKS

Ellipticity is deviation from a perfect circular form toward an ellipsoidal form. Measurement of the first stage nozzle is to determine the amount of deviation ("out-of-roundness") of the nozzle assembly when it is measured out of the unit. Ellipticity can be measured on the first stage nozzle since the segments are held in place by a retaining ring and are removed as an assembled unit in both the upper and lower halves. The second and third stage nozzle segments are assembled to the turbine casing and ellipticity checks are not required. Assistance from your GE Field Service Representative in obtaining ellipticity readings is recommended.

1. Set the upper and lower halves of the first stage nozzle on a reasonably level surface and take ellipticity readings as called for on Inspection Form GT9250.

REASSEMBLY

OPERATION 1 - INSTALL LOWER HALF FIRST STAGE NOZZLE AND ECCENTRIC PIN (WITH SUPPORT PAD AND RETAINER)

1. Thread a shouldered eyebolt into the center bolt hole in each of the nozzle horizontal joint flanges (one eyebolt for each end of the nozzle). These will be needed after the nozzle half is partially installed.
2. Using cable slings and chainfalls attached to the lifting bars on the back of the nozzle assembly, rig to lift the nozzle half inverted (ends pointed down), over the turbine casing. The lower half of the first stage nozzle weighs approximately 380 pounds. Check to ensure that the support pad seal ring groove in the first stage nozzle support ring is clean and there are no burrs on the tongue or in the groove prior to rolling in lower half first stage nozzle.
3. If the unit rotor is in place, begin rolling the nozzle half assembly into the turbine casing by letting out on one chainfall and taking up on the other, moving the crane as necessary to lead the assembly around the rotor.
4. Continue the roll-in process by talking up on the leading end cable and letting out on the following end cable.
5. By the time the nozzle half horizontal joints are oriented in the horizontal plane, both cable slings will be attached to the eyebolts previously installed in the horizontal joint bolt holes. With the chainfalls attached to the cable slings, lift the first stage nozzle half slightly and install the horizontal support blocks under the flanges.

NOTE Be sure that the horizontal joint support blocks are replaced in their original positions. These blocks are ground at factory assembly of the turbine to obtain the proper clearances for first-stage nozzle concentricity.

6. Lower the first stage nozzle half onto the horizontal joint blocks and remove the lifting slings, chainfalls and eyebolts. Ensure that the support pad seal joints are staggered at least 1.43 inches to 1.45 inches from the first stage nozzle segment joints.

7. Assemble the horizontal joint clamps, bolt then securely and record the clearances. Record the nozzle-to-rotor clearance on Inspection Form AES/GT-FF-6105. Refer to the Unit Clearance Diagram in the Reference Drawing section of the Service Manual for clearance specifications. It is recommended that all turbine clearances taken should be submitted to your local GE Field Service Representative prior to reassembly of turbine casing.

8. The eccentric pin is to be fitted to the first stage nozzle lower half after the nozzle has been assembled and properly positioned but before the upper half casing is installed. If the nozzle was not replaced and no adjustments were made to achieve proper clearance, the eccentric pin should seat properly without any adjustments. However if the nozzle was replaced or adjusted, the following procedure is to be used.

   a. Cut the old retaining strap from the top of the eccentric pin heads and grind both surfaces smooth.

   b. Install the eccentric pin into the casing to its full depth carefully as not to disturb the lateral centering of the nozzles.

   c. Coat top portion of the pin with Dyken Blue.

   d. Coat the top and sides of the retaining strap with Dyken Blue.

   e. Place the strap, coated side up, on top of the eccentric pin and bolt up with two bolting strap bolts.

      NOTE Apply a light coat of anti-seize compound to the bolt threads. Do not use colloidal copper type anti-seize.

   f. Scribe a mark on the heads of the pin along both sides of the retaining strap.

   g. Scribe two transverse marks across the tops and down the sides of the retaining strap and continuing across the head of the pin.

   h. Remove the retaining strap bolts, strap, and eccentric pin.

   i. Hold the pin in a vise, reposition the strap on the pin head, aligning the scribe marks, and weld the strap to the pin head as shown in Figure HGP-R1.

   j. Lightly coat the eccentric pin with anti-seize compound and reinstall it in the shell.

k. Install and properly tighten the retaining strap bolts.

OPERATION 1 - INSTALL LOWER HALF FIRST STAGE NOZZLE AND ECCENTRIC PIN (WITH TANGENTIAL SUPPORT PIN)

The following reassembly operation is for reinstalling the same first-stage nozzle that was removed from the unit. If a new or remanufactured nozzle is to be installed it is recommended that you obtain assistance from the GE Field Service Representative to install and properly align the first-stage nozzle.

1. Remove the eighteen 500-13 wire lock inserts (Helicoils) located in the retainer plate bolt hole in the support ring upper and lower halves and install new wire insert.
2. Thread a shouldered eyebolt into the center bolt hole in each of the nozzle horizontal joint flanges (one eyebolt for each end of the nozzle). These will be needed after the nozzle half is partially installed.
3. Using cable slings and chainfalls attached to the lifting bars on the back of the nozzle assembly, rig to lift the nozzle half inverted (ends pointed down), over the turbine casing. The lower half of the first stage nozzle weighs approximately 380 pounds.



Check to ensure that the groove in the first stage nozzle support ring is clean and there are no burrs on the tongue or in the groove prior to rolling in lower half first stage nozzle.

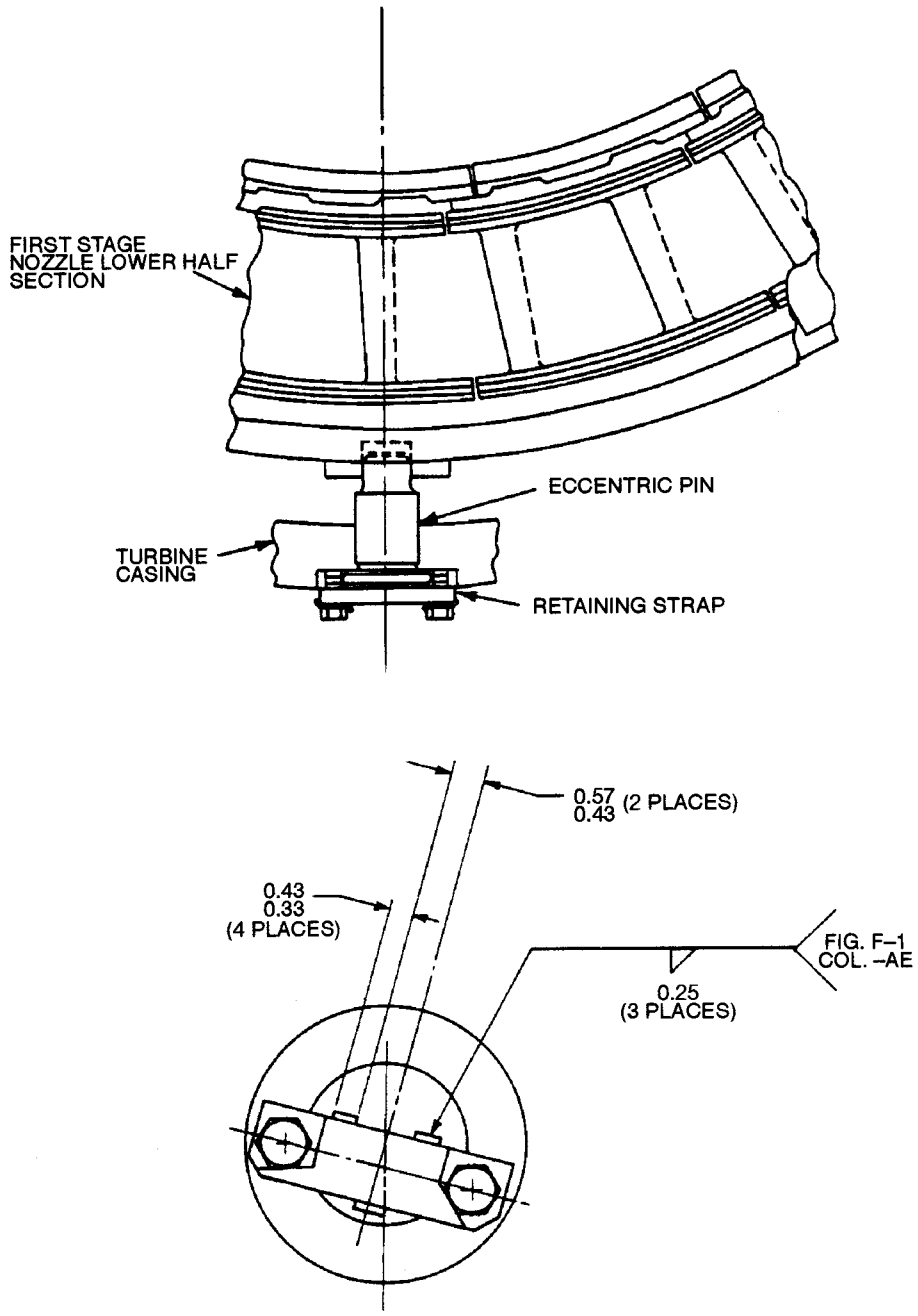


Figure HGP-R1. Eccentric Pin Retaining Strap Weld

4. If the unit rotor is in place, begin rolling the nozzle half assembly into the turbine casing by letting out on one chainfall and taking up on the other, moving the crane as necessary to lead the assembly around the rotor.

5. Continue the roll-in process by taking up on the leading end cable and letting out on the following end cable.

6. By the time the nozzle half horizontal joints are oriented in the horizontal plane, both cable slings will be attached to the eyebolts previously installed in the horizontal joint bolt holes. With the chainfalls attached to the cable slings, lift the first stage nozzle half slightly and install the horizontal support blocks under the flanges.

   NOTE Ensure that the horizontal joint support blocks are replaced in their original positions. These blocks are ground at factory assembly of the turbine to obtain the proper clearances for first-stage nozzle concentricity.

7. Lower the first stage nozzle half onto the horizontal joint blocks and remove the lifting slings, chainfalls and eyebolts.

8. Apply a light coating of Fel Pro C-102 anti-seize compound to the tangential support pins and install the pins into the support ring and first-stage nozzle support lugs.

9. Assemble the support pin retainer plate, coat the retainer plates bolts with anti-seize compound, install and tighten the bolts. Bend the locking tabs on the retainer plates to keep the bolts from rotating. See Figure HG P-R2

10. Assemble the horizontal joint clamps and bolt them securely. Record the clearances. Record the nozzle-to-rotor clearance on Inspection Form GT7200. Refer to the Unit Clearance Diagram in the Reference Drawing section of the Service Manual for clearance specifications. It is recommended that all turbine clearances taken be submitted to the GE Field Service Representative prior to reassembly of turbine casing.

11. The eccentric pin is to be fitted to the first stage nozzle lower half after the nozzle has been assembled and properly positioned but before the upper half casing is installed. If the nozzle was not replaced and no adjustments were made to achieve proper clearance, the eccentric pin should seat properly without any adjustments. However, if the nozzle was replaced or adjusted, the following procedure is to be used.

    a. Cut the old retaining strap from the top of the eccentric pin heads and grind both surfaces smooth.

    b. Install the eccentric pin into the casing to its full depth carefully as not to disturb the lateral centering of the nozzles.

    c. Coat top portion of the pin with Dyken Blue.

    d. Coat the top and sides of the retaining strap with Dyken Blue.

e. Place the strap, coated side up, on top of the eccentric pin and bolt up with two bolting strap bolts.

NOTE Apply a light coat of anti-seize compound to the bolt threads. Do not use a colloidal copper type anti-seize compound.

f. Scribe a mark on the heads of the pin along both sides of the retaining strap.

g. Scribe two transverse marks across the tops and down the sides of the retaining strap and continuing across the head of the pin.

h. Remove the retaining strap bolts, strap, and eccentric pin.

i. Holding the pin in a vise, reposition the strap on the pin head aligning the scribe marks, and weld the strap to the pin head as shown in Figure HGP-R1.

j. Lightly coat the eccentric pin with anti-seize compound and reinstall it in the shell.

k. Install and properly tighten the retaining strap bolts.

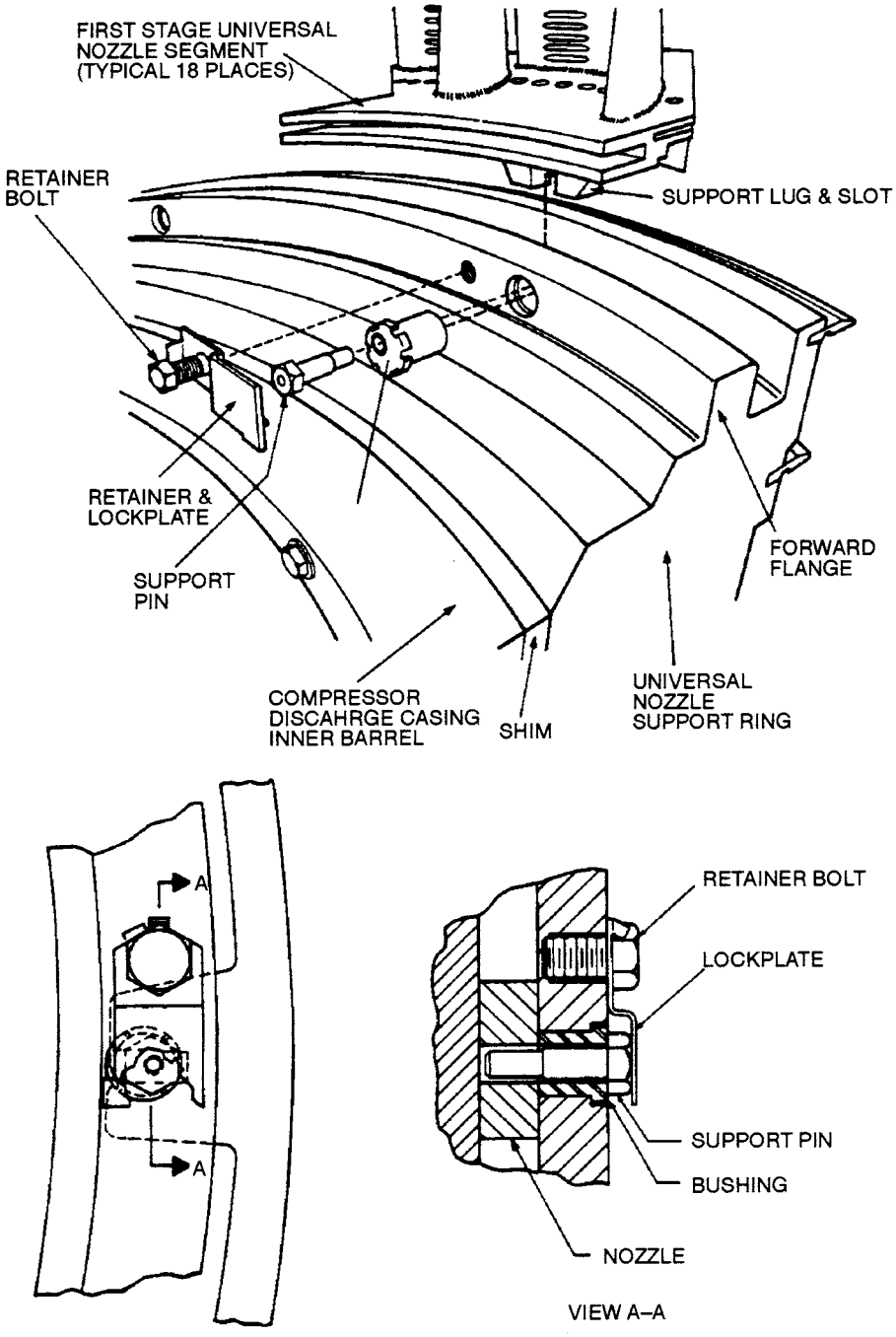


Figure HGP-R2. First Stage Nozzle Tangential Support Pin and Hardware Assembly

OPERATION 2 - INSTALL LOWER HALF SECOND AND THIRD STAGE NOZZLE SEGMENTS AND RADIAL RETAINING PINS

This operation assumes that the diaphragm and nozzle segments are being installed as one assembly.

1. Thoroughly clean the hook fits. Buffing the nozzle hook fits may be helpful prior to applying anti-seize compound. Apply a light coat of anti-seize compound.

   Use of an anti-seize compound must be restricted to a thin film such that the compound is not forced into the inner gas path.

2. Install the nozzle forward and aft seals into the nozzle segment in accordance with the Second and Third Stage Nozzle Arrangement drawing.

   Do not stake the forward and aft seals at the horizontal joint.

3. Ensure that the forward seals are bottomed out and that the ends of the aft seals are in contact with the forward seals. Assemble the axial seal strip with the notch facing out. Then assemble the radial seal strip with the notch facing in. Appropriately stake both seals. See Figures HGP-R3 and HGP-R4.

4. Lift the nozzle and diaphragm segment on its hook fit and slide it along to its proper position. Install nozzle segments from each horizontal joint to the center of the casing. For those nozzle segments with thermocouple tubes protruding, ensure that enough clearance is available to allow the segments to be rolled in without damaging the tubes. Each segment and diaphragm weighs approximately 35 pounds. Install nozzle segment radial locating pins after each segment is in position.

5. Apply a light coat of anti-seize compound to all radial retaining pins.

   Before installing the radial retaining pin threaded access plugs, clean the plug threads in the retaining pin hole to facilitate reassembly of the access plug.

6. Install the retaining pins as each nozzle segment is installed. Where plugs are used to hold the retaining pins, install the access plug so that it just bottoms on the retaining pin. Tighten the access plug against the retaining pin and stake each plug at four (4) locations on the edge of the threads.

7. Where pins are secured in place by holding plates, use lockplates to lock the hex head bolts.

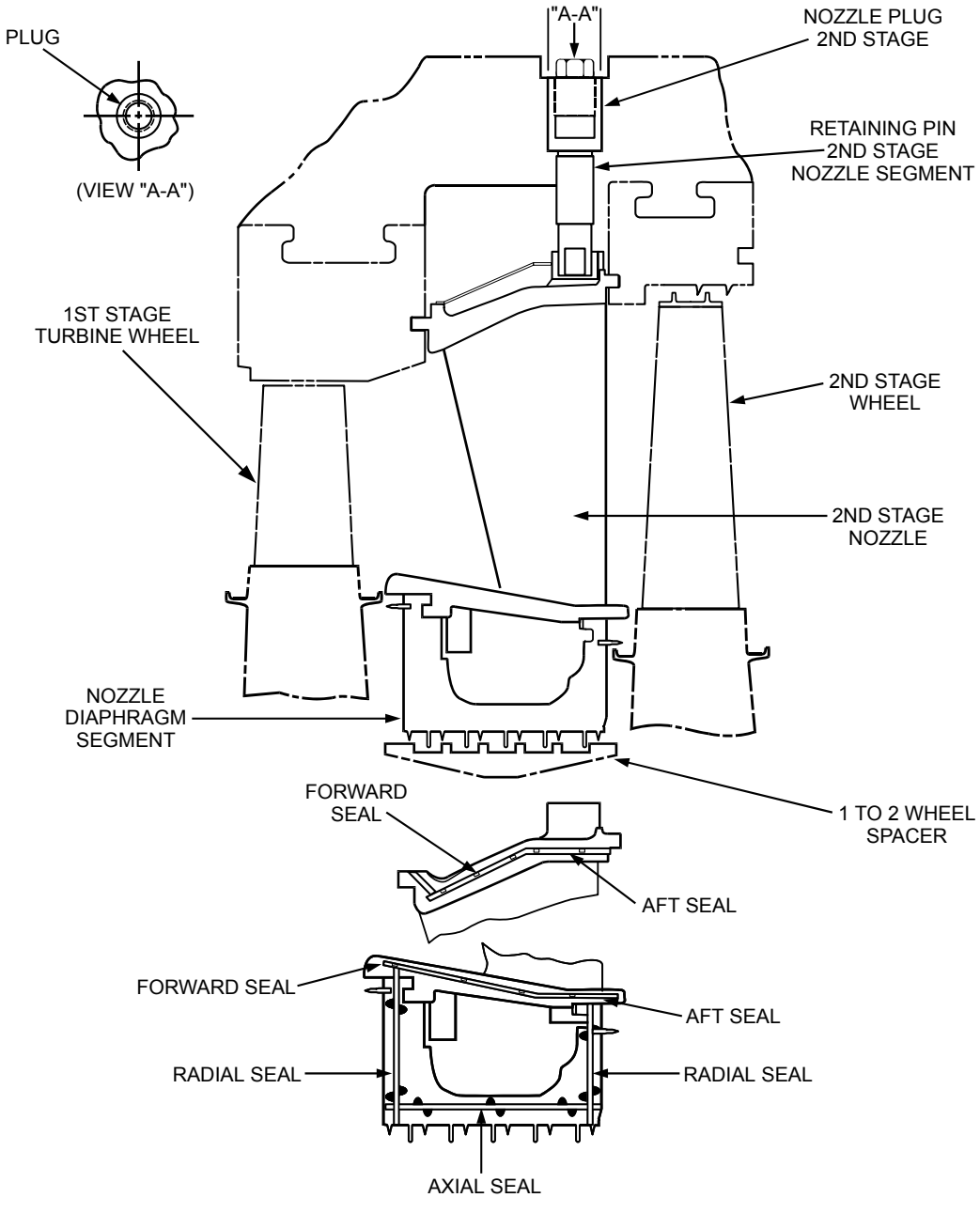


Figure HGP-R3. Second Stage Nozzle Seals and Radial Retaining Pins

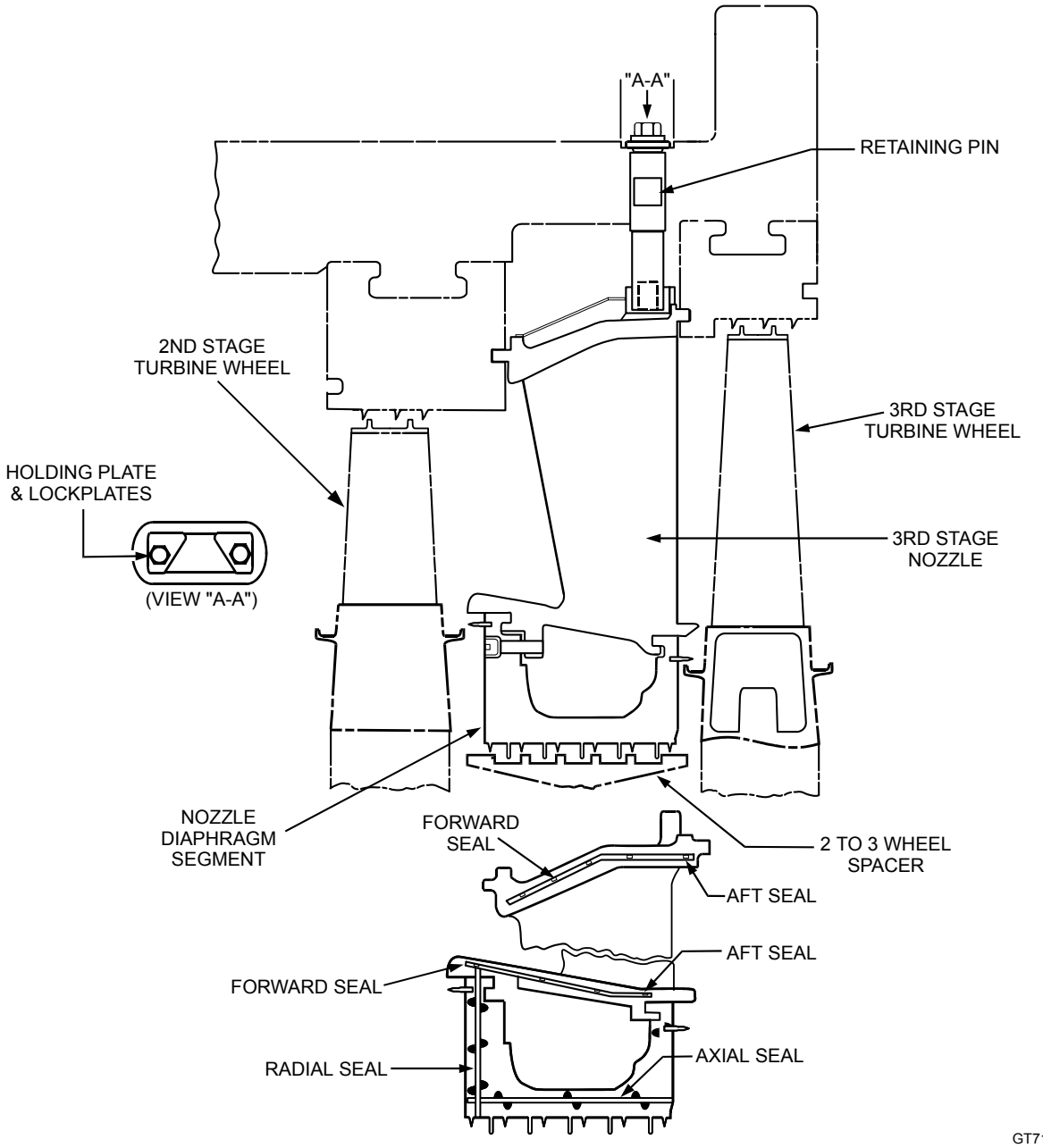


Figure HGP-R4. Third Stage Nozzle Seals and Radial Retaining Pins

OPERATION 3 - INSTALL WHEELSPACE THERMOCOUPLES IN LOWER HALF SECOND AND THIRD STAGE NOZZLE SEGMENTS

The removable type thermocouples are located in segments 10 and 15 in the second stage nozzle, and in segments 11 and 14 in the third stage nozzle. There are two thermocouples per segment. Refer to the second and third stage nozzle arrangement drawings in the Parts Lists and Drawings Volume of the Service Manual for details.

1. Prior to installing the wheelspace thermocouples, test the thermocouples to ensure they are operational. This can be done by applying a small amount of heat to the tip and reading the output on a temperature meter.

   NOTE The assembled length of the engaged portion of the thermocouple is stamped on the turbine casing adjacent to the thermocouple tubing installation hole. This dimension may be used for reference when installing the thermocouples. The reference dimension is from the top swagelok ferrell to the tip of the thermocouple.

2. Install the wheelspace thermocouples through the installation tubing in the turbine casing into the second stage nozzle segment numbers 10 and 15 (two per segment).

3. Install the wheelspace thermocouples through the installation tubing in the turbine casing into the third stage nozzle segment numbers 11 and 14 (two per segment).

OPERATION 4 - INSTALL UPPER HALF FIRST STAGE NOZZLE (WITH SUPPORT PAD AND RETAINER)

1. Check that the support pad groove in the first stage nozzle support ring is clean and free from burrs on the tongue or in the groove. See Figure HGP-R5.The upper half first stage nozzle assembly weighs approximately 380 pounds.

2. Rig and lift the upper half first stage nozzle assembly (ends pointing down) and carefully lower it onto the upper half support ring and lower half first stage nozzle.

3. Install and tighten the horizontal joint bolts.

OPERATION 5 - INSTALL UPPER HALF FIRST STAGE NOZZLE (WITH TANGENTIAL SUPPORT PIN)

1. Ensure that the groove in the first stage nozzle support ring is clean and that there are no burrs on the tongue or in the groove prior to installing the upper half first-stage nozzle. See Figure HGP-R5.

2. Rig and lift the upper half first-stage nozzle assembly (ends pointing down) and lower it onto the upper half support ring.

3. Apply a light coat of Fel-Pro C-102 anti-seize compound to the tangential support pins, and install the pins into the support ring and first stage nozzle support lugs.

4. Assemble the support pin retainer plate and coat the retainer plate bolt with anti-seize compound. Install and tighten the bolt. Bend the locking tabs on the retainer plate to keep the bolt from rotating. See Figure HGP-R2.

5. Install and tighten the first-stage nozzle horizontal joint bolts.

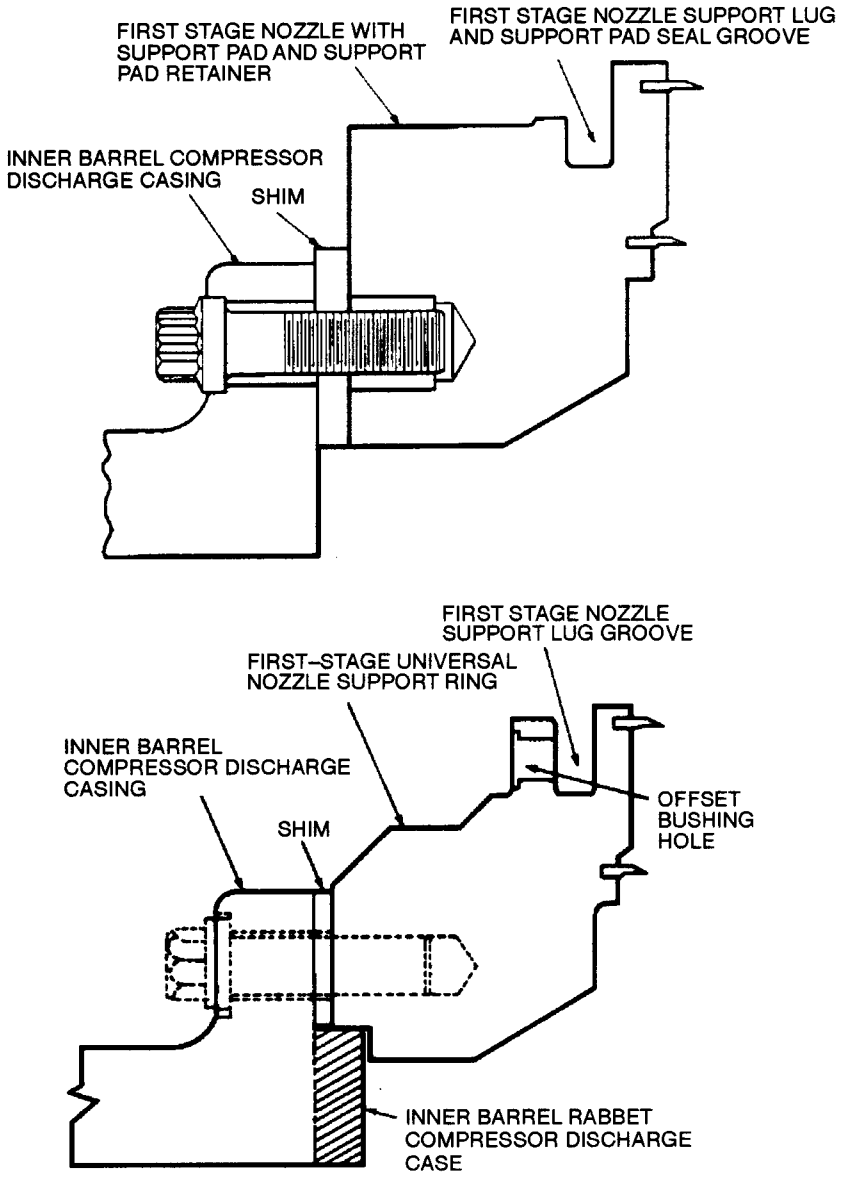


Figure HGP-R5. First Stage Nozzle Support Ring with Support Pad or Tangential Pin Groove

OPERATION 6 - INSTALL UPPER HALF SECOND AND THIRD STAGE NOZZLE SEGMENTS AND RADIAL RETAINING PINS

This operation assumes that the diaphragm and nozzle segments are being installed as one assembly.

1. Thoroughly clean the hook fits. Buffing the nozzle hook fits may be helpful prior to applying anti-seize compound. Apply a light coat of FEL-PRO C102 anti-seize compound to the hook fits.



Use of anti-seize compound must be restricted to a thin film such that the compound is not forced into the inner gas path. Should this occur, it must be carefully removed.

2. Install the nozzle forward and aft seals to the nozzle segment in accordance with the Nozzle Arrangement Drawings in the Parts Lists and Drawings Volume of the Service Manual.



Do not stake the forward and aft seals at the horizontal joint.

3. Ensure that the forward seals are bottomed out and that the ends of the aft seals are in contact with the forward seals.
4. Assemble the axial seal strip with the notch facing out. Then assemble the radial seal strip with the notch facing in. Appropriately stake both seals. See Figures HGP-R3 and HGP-R4.
5. Lift the nozzle and diaphragm segment onto its hook fit and slide it along to its proper position. Install nozzle segments from each horizontal joint to the bottom centerline of the casing. Each nozzle segment and diaphragm weighs approximately 35 pounds. Nozzle segment radial retaining pins should be installed after each segment is moved into position.
6. Apply a light coat of anti-seize compound to all radial retaining pins.
7. Install the radial retaining pins as each segment is installed. Where plugs are used to hold the retaining pins, install the access plug so that it just bottoms on the retaining pin. Tighten the access plugs against the retaining pins and stake each plug at four (4) places on the edge of the threads.
8. Where pins are secured in place by holding plates, use lockplates to lock the hexhead bolts.

OPERATION 7 - RECHECK FIRST STAGE NOZZLE CONCENTRICITY

Assistance from the GE Field Service Representative is recommended when making concentricity checks.

1. Take the concentricity readings as shown on Inspection Form GT9250. Record all readings on the inspection form and report readings to the GE Field Service Representative for evaluation and disposition.



The first stage nozzle is to be concentric with the support ring forward outside diameter within a tolerance of 0.050 inch, taking into account any ellipticity in the first stage nozzle assembly. Refer to the Alignment Diagram in the Reference Drawing section of the Service Manual. The first stage nozzle alignment should not be changed without first consulting the GE Field Service Representative.

OPERATION 8 - RECHECK TURBINE CLEARANCES

1. All turbine clearances should be retaken per Hot Gas Path Inspection Disassembly Operation Number 8 and checked by the GE Field Service Representative prior to reassembly of the turbine casing upper half.
2. Record all final clearance readings on Inspection Form GT7200.

OPERATION 9 - INSTALL TRANSITION PIECES

Refer to Combustion Inspection Section 4A, Reassembly Operations 1 and 2, of the Service Manual for transition piece installation procedure.

OPERATION 10 - INSTALL UPPER HALF TURBINE CASING AND UPPER HALF FIRST STAGE NOZZLE ECCENTRIC PIN

1. Attach and rig wire rope slings to the outer upper lugs of the spreader bar and to the overhead crane. Lift the spreader bar and position it over the turbine casing.
2. Using wire rope slings, rig the two aft lower lugs of the spreader bar to the aft lugs of the turbine casing. Using chainfalls or come-a-longs and wire rope slings, rig the forward lower set of lugs to the forward lugs on the turbine casing. Position the spreader bar as near to the turbine casing as a safe lifting angle will allow. Rigging should be such that when the wire rope slings are taut, the bottom of the spreader bar should be fairly close to the highest section of the turbine casing.



The upper half turbine casing weight with the second and third stage nozzle segments and shrouds weighs approximately 7300 pounds.

3. Flat file and wipe off the horizontal joints of the turbine casing lower half.

4. Apply Perfect Seal to the surface of the horizontal joint of the turbine case lower half and the vertical joints of the compressor discharge casing and exhaust frame for a distance of one inch (1) on both sides of the four way joints.



Apply only a thin film of anti-seize compound or it will interfere with pulling the joint together and act as a gasket. Do not allow anti-seize compound to get into the hot gas path areas.

5. Apply a thin film of anti-seize compound to the horizontal joint of the turbine casing lower half, and to the vertical joints of the compressor discharge casing and exhaust frame.



Check the seals and seal strip fits for second and third stage nozzle segments. It may be necessary to grind them to facilitate realignment and reassembly of the upper half nozzles and turbine casing to the lower half turbine casing and nozzle segments.

6. Lift the turbine casing and level it. Flat file and wipe off the horizontal and vertical joints.

7. Lightly lubricate and install guide pins in the upper half casing horizontal joints bodybound hole locations.

8. Lift the turbine casing over the center of the unit and lower the casing until the aft end clears the lifting beam and trolley hoist. Move the trolley hoist up against its forward stop.

9. Using the overhead crane, move the turbine casing axially aft and rig the trolley hoist to the center lug on the spreader bar. Using the trolley hoist, take up the rigging so as to transfer the load to the trolley hoist. Remove the overhead crane rigging. See Figure HGP-R6.

10. Inspect the lower half turbine casing to ensure all items are secure and for foreign objects.

11. Position the upper half turbine casing directly over the lower half turbine casing with the trolley hoist.



Do not allow the turbine casing to swing and hit the turbine buckets while it is being lowered into position.

12. Lower the turbine casing. Using the guide pins provided, ensure the horizontal joint sealing strips at the second and third stage joints are in position.



Ensure that the upper and lower half turbine nozzle segments are loaded in the same direction to avoid mismatch at the horizontal joint seals.

13. Coat the top eccentric pin with anti-seize compound and insert it into the casing. Do not force in the pin. Refer to Reassembly Operation 1 if the eccentric pin needs to be reset.

14. Assemble the eccentric pin locking bar bolts to the casing and secure them with locking tabs.

15. Install all bolts hand-tight, including bodybound bolts. Tighten the horizontal joint bolts, starting at the mid-point and working alternately left and right forward and aft until all bolts are tightened.

16. Alternately tighten all the vertical bolts, working down each side from the top vertical centerline to the horizontal joint. See Figure HGP-R7. Work both the aft and forward vertical flanges simultaneously.

17. It may be necessary to remove the horizontal joint bolts next to the vertical joint in order to tighten the vertical joint bolts. Replace the horizontal joint bolts and tighten.

18. Using chainfall or come-a-long, rig to the rail beam and trolley hoist inside the exhaust transition duct. Remove the bolting from the rail beam and slide the rail and trolley out of the duct and rig the rail to the overhead crane. Remove rail and hoist clear of the unit. Install all removed mounting hardware hand-tight onto the beam for safe keeping.

OPERATION 11 - INSTALL WHEELSPACE THERMOCOUPLE WIRING AND CONDUIT

1. Reinstall conduit for wheelspace thermocouple wiring in same location from which it was removed. Use anti-seize compound on threaded conduit fittings and covers.

2. Check thermocouple identification lead tags and reconnect leads. Tape with glass tape.

NOTE

Refer to the Conduit Arrangement Wheelspace Thermocouples drawing in the Parts Lists and Drawings Volume of the Service Manual for wheelspace thermocouple identification.

OPERATION 12 - REMOVE MECHANICAL SUPPORT JACKS FROM UNDERNEATH TURBINE CASING

Caution

Do not remove mechanical support jacks until the casing has been installed and bolted.

1. Remove the mechanical support jacks and dial indicators from beneath the unit casing flanges.

2. Remove shim packs if used (applicable to bases with a transaxial "I" beam only).

3. Replace the base cutout floor plates and bolt in place.

4. Gather all jacks and dial indicators and return to storage.

OPERATION 13 - TAKE FINAL COMPRESSOR AND TURBINE ROTOR POSITIONING CHECKS

Submit all readings from both the initial and final rotor positioning checks to the GE Field Service Representative for evaluation. All final readings must be within tolerances as specified on the Clearance Diagram in the Reference Drawing section of the Service Manual.

1. To take final rotor positioning checks, follow procedures outlined in Disassembly Operation 3.
2. Record all readings on Inspection Form GT7200, Rotor Positioning Checks.

OPERATION 14 - PERFORM COMBUSTION INSPECTION REASSEMBLY OPERATIONS 4 THROUGH 19

Refer to Combustion Inspection Section 4A in the Service Manual for reassembly procedures for the following operations.

1. Install outer combustion casings and aft flow sleeves.
2. Install inner crossfire tube assemblies and check clearances.
3. Install outer crossfire tubes, adapter plates, gaskets, ring packings and packing glands.
4. Install forward flow sleeves, inner crossfire tubes, combustion liners and spring retainers.
5. Assemble flame detectors and spark plugs.
6. Close and bolt combustion casing hinged covers.
7. Reconnect false start drain manifold and drain lines to combustion casing.
8. Install 11th stage cooling and sealing air lines.Install all the 11th stage cooling and sealing air piping except the upper right side section with the compressor bleed valve (33CB-2). This section will be installed after the exhaust plenum access upper panel is reinstalled.
9. Install fuel nozzles.
10. Install fuel check valves.
11. Install liquid fuel tubing lines.
12. Install water injection tubing pigtails (if applicable).
13. Install atomizing air lines.
14. Install fuel gas piping (if applicable).
15. Install steam injection piping pigtails (if applicable).

16. Install the atomizing air extraction manifold to upper half turbine casing. In addition, reinstall the inlet deicing piping, if applicable, and the exhaust frame cooling piping. Install new gaskets, coat bolts with Fel-P C-102 anti-seize compound, install bolting and tighten using a crisscross sequence.

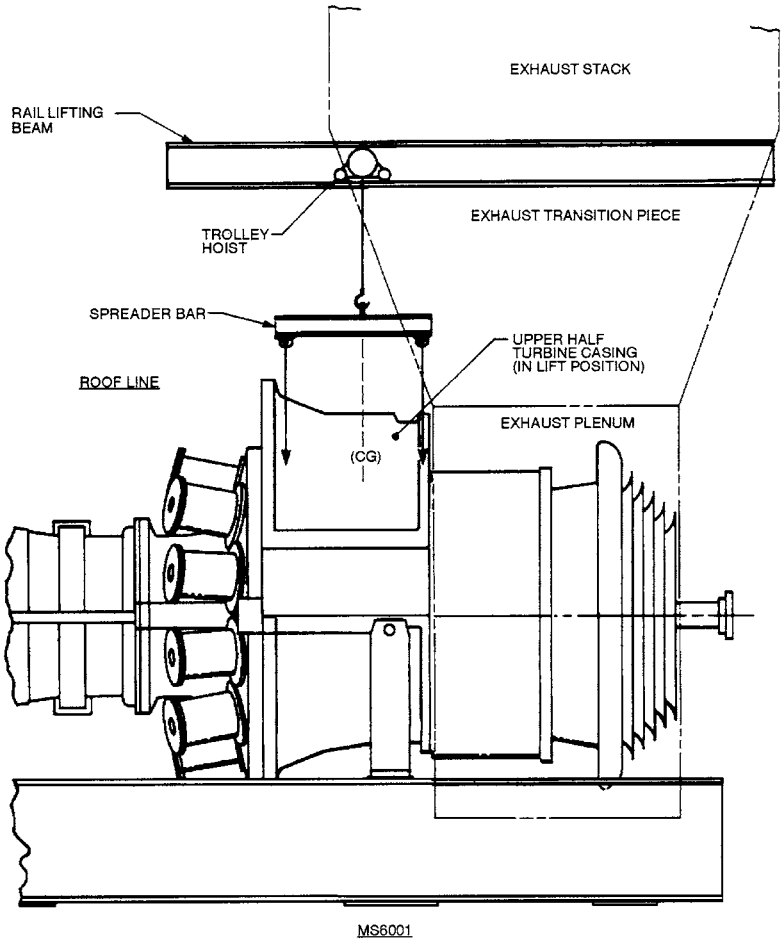


Figure HGP-R6. Turbine Casing Rigged for Reinstallation

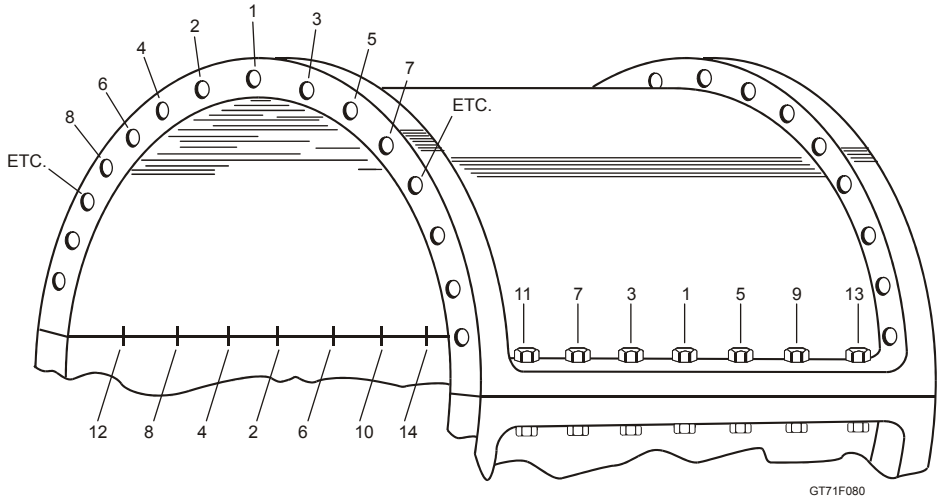


Figure HGP-R7. Typical Bolt Tightening Sequence for Unit Casings

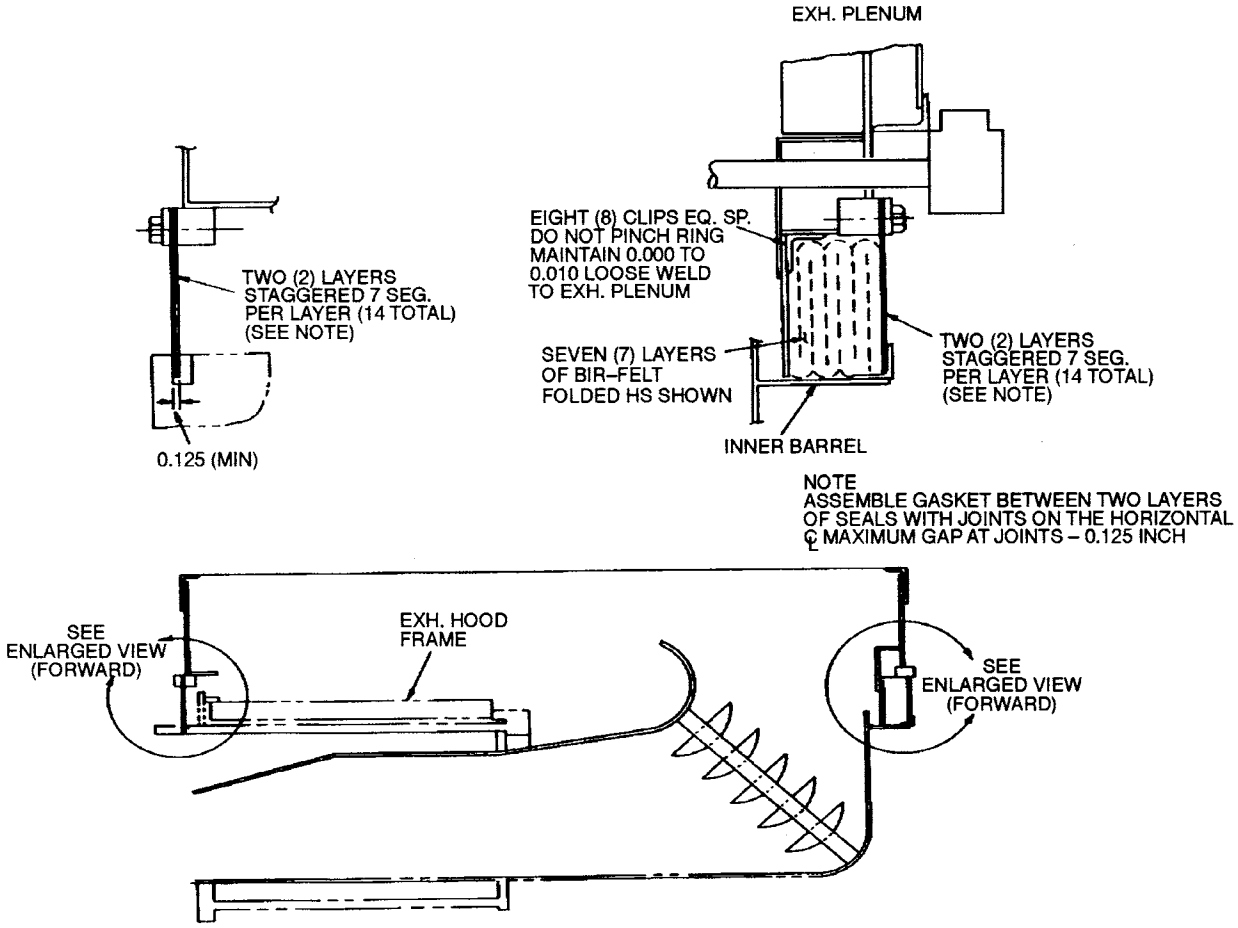


Figure HGP-R8. Exhaust Seal Segments, Forward and Aft Sides

OPERATION 15 - REASSEMBLE EXHAUST AND INLET DUCT ACCESS PANELS

1. On a straight up exhaust configuration, spread the exhaust plenum frame using a length of 4 inch pipe or timber and a 10-ton mechanical jack to allow sufficient clearance to install the access panel.
2. Assemble new gasket, lift and position the panel into the plenum opening. Coat bolting with Fel-Pro C-102 anti-seize compound. Install bolts but do not tighten.
3. Assemble new gasket to the exhaust duct transition piece forward panel. Lift and guide the panel into position. Come-a-longs will be required to help guide and position the panel into location underneath the overhang of the duct transition piece.
4. Coat bolts with Fel-Pro C-102 anti-seize compound. Install bolts and tighten. Remove rigging.
5. Tighten the bolts on the exhaust plenum panel.
6. Reassemble the exhaust plenum forward flex seals gaskets and clamp bars between the exhaust hood frame and plenum. See Figure HGP-R8.
7. Reinstall the 11th stage extraction piping and compressor bleed valve to the exhaust plenum panel. Reconnect all tubing runs and piping runs to the 11th stage piping and bleed valve.
8. Inspect the inlet plenum for cleanliness and or foreign objects. Install new gasket on the inlet plenum access panel located in the lower right hand section of the plenum (looking **with** air flow). Coat bolts with anti-seize compound. Install access panel and bolt in place.

OPERATION 16 - ASSEMBLE TURBINE COMPARTMENT ROOF, SIDE PANELS AND DOORS

1. Reassemble lagging side panel vertical beams if they were removed at disassembly.
2. Attach four (4) eyebolts to threaded receptacles on the roof. Rig the roof (include a chainfall in rigging) for lifting.
3. Apply new gasketing or caulking to roof section.
4. Using a chainfall, lift the roof level over the turbine compartment and carefully lower it into position.
5. Apply Fel-Pro C-102 anti-seize compound to the bolts and bolt the roof in place.
6. Reconnect blower fan motor leads and conduit.
7. Reconnect exhaust frame cooling piping flanges and tubing connection.
8. Reconnect VA14 valve, wiring and conduit (if applicable).

9. Make a complete visual inspection of roof topside and underside to ensure all components have been connected and assembled.

10. Install turbine compartment lagging panels and doors.

OPERATION 17 - CLEAN, OPERATE AND LEAK CHECK UNIT

1. Clean up all fuel oil, lube oil and water spills.

2. Visually inspect the unit. Check to ensure that the following has been completed.

   a. Check that all removed piping has been reinstalled.

   b. Check that all removed conduit has been reinstalled and that electrical connections have been made.

   c. Confirm proper functioning of all devices.

   d. Correct any fuel, oil and water leaks.

   e. Inspect inlet and exhaust plenums for cleanliness and secure the unit.

3. Reconnect power to all systems that were disabled at the beginning of the job.

4. Rearm the fire protection system and manually reset and latch the CO2 vent dampeners.



It is recommended that the following checks be made under the guidance of the GE Field Service Representative.

5. Perform pre-cranking checks as follows.

   a. Record all panel counter readings.

   b. Check operation of the cooldown/emergency lube oil pump.

   c. Check operation of auxiliary hydraulic pump.

   d. Perform visual inspection to determine if oil is flowing in bearing drains.

   e. Check atomizing air compressor.

   f. Block out excitation and generator breaker equipment.

   g. Place unit on ratchet and observe for leaks, abnormal noises, or control deficiencies.

   h. Prime the fuel pump.

6. Crank the unit and perform the following checks.

   a. Observe pickup RPM of speed sensors vs. time.

   b. Inspect casing for air leaks.

   c. Observe and record vibration levels.

d. Observe and record maximum crank speed.

e. Check starting means for proper operation.

f. Listen for abnormal noises.

g. Check temperature of lube oil in bearing drains or at the bearing header and tank.

h. Observe all panel pressure gauge readings for abnormalities.

i. Trip the unit using the emergency trip. Listen for abnormal noises from the gas turbine and associated equipment during coastdown.

j. Observe proper drop-out RPM of speed sensors.

7. Initiate a start and perform the following zero-to-full-speed-no-load checks.

   a. Observe applicable firing check parameters.

   b. Check spark plugs and flame detectors.

   c. Observe and record acceleration rates.

   d. Record full set of vibration readings.

   e. Observe and record temperature suppression setpoint.

   f. Note and record pick up time of all appropriate electrical speed relays.

   g. Adjust full-speed-no-load RPM to specifications.

   h. Check digital setpoint.

   i. Take a complete set of running data and record on Operating Data form as shown in Running Inspection, Section 4 of the Service Manual.

   j. Perform governor checks.

   k. Perform overspeed checks per unit control specifications in unit Service Manual.

   l. After shutdown, stake sight port plugs in place using two stakes per plug.

   m. Periodically recheck torque on fuel nozzle mounting flange bolts after shutdown.

8. After synchronizing the unit and closing the breaker, perform the following load checks (as applicable).

   a. Take full set of load data per Steps 1, 2 and 3 below and record on Operating Data form as shown in Running Inspection section.

      1. Preselect load points.

      2. Base load.

      3. Peak load.

   b. Check for air, oil, fuel, gas and water leaks.

9. Observe normal unloading and shutdown as follows.
   a. Note and record coastdown time.
   b. Listen for abnormal noises from the gas turbine and associated equipment during coastdown.
   c. Observe unit vibration readings during coastdown.
   d. Check to ensure sight port pipe plugs have been staked in two places after shutdown.
   e. Periodically recheck torque on fuel nozzle mounting flange bolts after shutdown.
10. Inspection Report
   a. Submit a completed FSRAuto Report using the Mail Function within FSRAuto. The report should contain all electronic inspection forms attached within the report, and should be completed and turned into the regional office one week after outage completion.

REVISION PAGE

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June 11, 2003

Revision Date	Revision	Revision Details	Approved By
3/31/2004	A	Added reference document EM 5202 – Gas Turbine Mechanical Support and Jacking Instructions	MJC